SIMULTANEOUS RECOVERY OF ELASTIC AND VISCOUS COEFFICIENTS IN A KELVIN–VOIGT EQUATION

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Abstract. We study the simultaneous recovery of a spatially dependent elastic coefficient and a time- and space-dependent viscosity coefficient in a scalar Kelvin–Voigt equation. We prove that the displacement-to-traction map over any fixed time interval uniquely determines both coefficients, provided that the Riemannian metric induced by the viscosity is simple at each time. In the velocity formulation, the elastic term gives rise to a Volterra memory term involving second-order spatial derivatives, with a corresponding memory contribution to the boundary traction. We first determine all normal derivatives of both coefficients at the boundary using highly oscillatory boundary data. We then construct exact parabolic exponential solutions in the presence of memory and determine the viscosity by boundary rigidity. With the viscosity determined, an approximate identity argument reduces recovery of the elastic coefficient to injectivity of the scalar geodesic ray transform. Finally, numerical experiments illustrate a regularized finite-dimensional reconstruction of both coefficients from finitely many moments of the displacement-to-traction map.

1. Introduction

Viscoelastic solids exhibit both a recoverable deformation response and energy dissipation during motion. The Kelvin–Voigt constitutive law models these effects by adding a viscous stress proportional to the strain rate to an elastic stress proportional to the strain [Chr82, SO20]. This rate-dependent formulation has been used to study material damping in biomimetic scale-covered beams [EKAG23]. Kelvin–Voigt elements have also been used to represent the viscoelastic membrane and cytoplasm within a hierarchical model of cellular rheology [HKXG21].

Motivated by applications to heterogeneous materials, we allow the elastic coefficient to depend on position and the viscosity to depend on both position and time. In a scalar displacement approximation, the resulting stress flux is $a(x)\nabla u + b(t,x)\nabla\partial_t u$. Combining the divergence of this flux with the balance of linear momentum, and normalizing the mass density to one, yields the equation (1.1). Physically, predicting the response of a heterogeneous viscoelastic body requires knowledge of its internal stiffness and damping. These parameters are often difficult to measure directly, whereas boundary displacements can be prescribed and the resulting tractions observed. This motivates us to study the simultaneous recovery of these coefficients from suitable boundary measurements.

Let $T > 0$ and let $\Omega \subset \mathbb{R}^n$, $n \geq 2$, be a connected bounded domain with smooth boundary $\partial\Omega = \Gamma$. Throughout this paper, we set

$$Q = (0,T) \times \Omega, \quad \Sigma = (0,T) \times \Gamma.$$

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We consider the following scalar Kelvin–Voigt equation

$$\begin{cases} \partial_t^2 u - \operatorname{div}(a(x)\nabla u) - \operatorname{div}(b(t,x)\nabla \partial_t u) = 0 & \text{in } Q, \\ u = f & \text{on } \Sigma, \\ u(0,\cdot) = \partial_t u(0,\cdot) = 0 & \text{in } \Omega. \end{cases} \tag{1.1}$$

Here u denotes the scalar displacement, $a(x)$ is the elastic coefficient and $b(t,x)$ is the time-dependent viscosity coefficient. The terms $a\nabla u$ and $b\nabla\partial_t u$ are the elastic and viscous stress fluxes, respectively. We note that the mass density of a given medium has been normalized to one.

Henceforth, we assume that the coefficients $a \in C^\infty(\overline{\Omega})$ and $b \in C^\infty([0,T]\times\overline{\Omega})$ are real-valued and that there exist positive constants a_*, b_* such that

$$a(x) \geq a_* > 0, \quad b(t,x) \geq b_* > 0. \tag{1.2}$$

1.1. **Formulation of the inverse problem and main result.** We next formulate the considered inverse problem. Define the stress flux by

$$\sigma(u) := a\nabla u + b\nabla\partial_t u.$$

For the moment, we formally introduce the displacement-to-traction map by

$$\Lambda_{a,b}^T f := \langle \sigma(u), \nu\rangle|_\Sigma = (a\partial_\nu u + b\partial_\nu\partial_t u)|_\Sigma. \tag{1.3}$$

Here, ν denotes the outward unit normal vector field of Ω along Γ. The well-posedness of the forward problem (1.1) is established in Proposition 2.2, so that the map $\Lambda_{a,b}^T f$ is well-defined. We note that, for real smooth solutions, we have the energy identity

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\int_\Omega (|\partial_t u|^2 + a|\nabla u|^2)\,\mathrm{d}x + \int_\Omega b|\nabla\partial_t u|^2\,\mathrm{d}x = \int_\Gamma \sigma(u)\cdot\nu\,\partial_t u\,\mathrm{d}S.$$

This provides a natural derivation of the displacement-to-traction map.

Let K_+ and K_- be the forward and backward Volterra operators,

$$(K_+h)(t) = \int_0^t h(s)\,\mathrm{d}s, \quad (K_-h)(t) = \int_t^T h(s)\,\mathrm{d}s.$$

We consider lateral Dirichlet data in the class

$$\mathcal{D}_T := K_+ C_c^\infty(\Sigma) = \{K_+h : h \in C_c^\infty(\Sigma)\}.$$

Thus, for $f \in \mathcal{D}_T$, the boundary velocity $h = \partial_t f$ is smooth and compactly supported in Σ. The displacement f vanishes near $t = 0$ but is constant, and need not vanish, near $t = T$. Section 2 establishes well-posedness for a larger Sobolev class of boundary data and the boundedness of the associated displacement-to-traction map.

The concerned inverse problem is stated as follows.

(IP) *Can one simultaneously recover the elastic coefficient $a(x)$ and the time-dependent viscosity $b(t,x)$ by the boundary displacement-to-traction map $\Lambda_{a,b}^T$?*

For fixed time t, the relevant Riemannian metric associated with b is given by

$$g_b(t) = b^{-1}(t,\cdot)e, \quad e = \mathrm{d}s^2 = \mathrm{d}x_1^2 + \cdots \mathrm{d}x_n^2, \tag{1.4}$$

where e denotes the standard Euclidean metric. A compact connected Riemannian manifold $(\mathcal{M}, g)$ with smooth boundary is *simple* if its boundary is strictly convex, it is nontrapping, and it has no conjugate points. In a simple manifold, any two points are joined by a unique geodesic, which is minimizing (see [PSU23, Section 3.8]).

Below, we give an affirmative answer to the considered inverse problem.

Theorem 1.1. *Let $T > 0$ be given. For $j = 1, 2$, let $a_j \in C^\infty(\overline{\Omega})$ and $b_j \in C^\infty([0,T] \times \overline{\Omega})$ be real-valued and satisfy the condition (1.2). Assume that the geometry $(\overline{\Omega}, g_{b_j}(t))$ is simple for every $t \in [0,T]$. If*

$$\Lambda^T_{a_1,b_1} f = \Lambda^T_{a_2,b_2} f \quad \text{for all } f \in \mathcal{D}_T, \tag{1.5}$$

then $a_1 = a_2$ in $\overline{\Omega}$ and $b_1 = b_2$ in $[0,T] \times \overline{\Omega}$.

Remark 1.2. Theorem 1.1 uses the displacement-to-traction map on the whole lateral boundary for all $f \in \mathcal{D}_T$, over any fixed $T > 0$. It requires neither prior boundary agreement of the coefficients nor time analyticity of b_j. Simplicity is required only for $g_{b_j}(t) = b_j^{-1}(t,\cdot)e$, with no geometric condition on $a_j^{-1}e$.

Remark 1.3. The positivity of a in (1.2) is assumed only for the physical interpretation of the energy. Mathematically, Theorem 1.1 holds for any real-valued $a \in C^\infty(\overline{\Omega})$.

1.2. **Related literature and main difficulties.** Inverse problems for dissipative evolution equations have attracted considerable attention over the past decades, motivated by applications such as parameter identification, boundary data completion, and source or coefficient determination, as well as by the mathematical challenges arising from the loss of time reversibility and the smoothing effect of the associated semigroup. In particular, strongly damped wave and Kelvin–Voigt equations have been studied with a variety of unknowns and measurement schemes. Carleman estimates have been used to obtain inverse source results for Kelvin–Voigt systems [IY19] and to recover a lower-order coefficient in a strongly damped wave equation [Wu12].

Li and Zhang [LZ24b] recovered the underlying metric up to isometry in a strongly damped wave equation whose elastic and damping terms involve the same Laplace–Beltrami operator. Related work in nonlinear acoustics addresses the recovery of a time-dependent potential and a nonlinear coefficient [LZ24a], as well as the simultaneous recovery of wave speed, diffusivity, and nonlinearity from boundary Cauchy data at the first and second harmonics under time-harmonic boundary excitation at a sufficiently high frequency [AP26]. Yamamoto, Zhang, and Zhao [YZZ26] established Hölder stability in recovering five spatially varying coefficients in a three-dimensional viscoelastic system, using finitely many measurements along with fixed-time interior data.

For equations with memory, boundary measurements have been used to recover memory kernels, including from partial boundary data [BDI97, BDU02], and lower-order coefficients in wave equations with nonlocal attenuation [KRU26]. Memory kernels in strongly damped wave equations have also been recovered from supplementary observations, including a flux condition [Col07, CG09]. In the parabolic setting, coefficient recovery from boundary measurements was studied in [CK01]. Carleman estimates have also yielded stability results for equations with second-order spatial memory [BSB08].

For the viscoacoustic wave equation, exterior source-to-solution data have been shown, under suitable geometric assumptions, to determine the sound speed and all time derivatives at zero of the kernel of a memory term involving second-order spatial derivatives [CdHS26]. High-frequency propagation in the presence of weakly singular hereditary memory has been studied in [GS26].

Our approach to recovering the time-dependent viscosity draws on the geometric approach to inverse boundary problems for nonautonomous heat equations developed in Feizmohammadi's work [Fei24]. In the velocity formulation of the Kelvin–Voigt equation, the elastic term gives rise to a Volterra memory term involving two spatial derivatives.

The boundary traction contains a corresponding memory contribution that must also be controlled in the inverse analysis.

We first construct boundary-localized, highly oscillatory solutions and use them to determine all normal derivatives of both coefficients at the boundary. We then construct exact parabolic exponential solutions by showing that the conjugated memory operator, composed with the inverse of the conjugated parabolic operator, is small as the exponential parameter tends to infinity. After controlling the memory contribution to the boundary traction, we recover the viscosity by boundary rigidity. Once the viscosity is known, a strong approximate-identity argument reduces the recovery of the elastic coefficient to the injectivity of the scalar geodesic ray transform applied to $(a_1 - a_2)/b(t, \cdot)$.

The remainder of the paper is organized as follows. In Section 2, we develop the parabolic formulation of the original equation, establish well-posedness using a global exponential weight, and derive two useful identities related to the recovery of coefficients. In Section 3, we study the boundary determination of all normal derivatives of both coefficients at the boundary. Section 4 focuses on constructing exact exponential solutions with geodesic distance functions as phases in the presence of the Volterra memory term. The proof of the main theorem will be provided in Section 5. Finally, Section 6 presents a finite-dimensional reconstruction method and numerical experiments.

2. The forward problem and boundary identities

In this section, we begin by reformulating the Kelvin–Voigt equation as a parabolic equation with a Volterra memory term, and then establish well-posedness and the boundary identities used in the inverse analysis. Let $v = \partial_t u$. Since $u(0, \cdot) = 0$, we have $u = K_+ v$, where K_+ acts in the time variable. Set

$$D_a = \operatorname{div}(a\nabla), \quad P_b^{\pm} = \pm\partial_t - \operatorname{div}(b\nabla), \quad L_{a,b}^{\pm} = P_b^{\pm} - K_{\pm} D_a.$$

As a is independent of time, $D_a K_+ v = K_+ D_a v$. Hence (1.1) is equivalent to

$$\begin{cases} L_{a,b}^{+} v = 0 & \text{in } Q, \\ v = h = \partial_t f & \text{on } \Sigma, \\ v(0, \cdot) = 0 & \text{in } \Omega, \end{cases} \qquad u = K_+ v. \tag{2.1}$$

2.1. Trace spaces and well-posedness results.

We introduce the following Hilbert spaces

$$\mathbb{Y} = L^2(Q), \qquad \mathbb{W} = H^1(0, T; L^2(\Omega)) \cap L^2(0, T; H^2(\Omega)),$$
$$\mathbb{X} = \mathbb{W} \cap L^2(0, T; H_0^1(\Omega)), \quad \mathbb{X}_+ = \{z \in \mathbb{X} : z(0) = 0\}, \quad \mathbb{X}_- = \{z \in \mathbb{X} : z(T) = 0\}.$$

The spaces $\mathbb{X}$ and $\mathbb{X}_\pm$ are equipped with the norm inherited from $\mathbb{W}$. The embedding $\mathbb{W} \hookrightarrow C([0, T]; H^1(\Omega))$ ensures that the endpoint conditions are well defined. The lateral Dirichlet and normal trace spaces are

$$\begin{aligned} \mathbb{H} &= H^{3/4}(0, T; L^2(\Gamma)) \cap L^2(0, T; H^{3/2}(\Gamma)), \\ \mathbb{H}_+ &= \{h \in \mathbb{H} : h(0) = 0\}, \quad \mathbb{H}_- = \{k \in \mathbb{H} : k(T) = 0\}, \\ \mathbb{N} &= H^{1/4}(0, T; L^2(\Gamma)) \cap L^2(0, T; H^{1/2}(\Gamma)). \end{aligned} \tag{2.2}$$

The endpoint conditions defining $\mathbb{H}_\pm$ are understood in $H^{1/2}(\Gamma)$. The joint initial–lateral trace theorem provides bounded liftings $E_\pm : \mathbb{H}_\pm \to \mathbb{W}$ with the prescribed lateral trace and zero initial or terminal trace, respectively. Moreover, for every $z \in \mathbb{W}$, we have

$$\|\partial_\nu z\|_{\mathbb{N}} \le C \|z\|_{\mathbb{W}}. \tag{2.3}$$

We will use the scalar Dirichlet maximal-regularity and trace results of [DHP07, Theorem 2.1 and Section 3]. The positivity and smoothness of b allow us to apply Dirichlet maximal regularity. For $\eta \in \mathbb{H}_{\pm}$, the lifting $E_{\pm}\eta \in \mathbb{W}$ also satisfies $L^{\pm}_{a,b}E_{\pm}\eta \in \mathbb{Y}$.

Proposition 2.1. *For any $F \in \mathbb{Y}$ and $h \in \mathbb{H}_+$, the initial-boundary value problem*

$$\begin{cases} L^{+}_{a,b}v = F & \text{in } Q, \\ v = h, & \text{on } \Sigma \\ v(0) = 0 & \text{in } \Omega, \end{cases} \tag{2.4}$$

admits a unique solution $v \in \mathbb{W}$, satisfying the estimate

$$\|v\|_{\mathbb{W}} + \|\partial_\nu v\|_{\mathbb{N}} \le C\Big(\|F\|_{\mathbb{Y}} + \|h\|_{\mathbb{H}}\Big). \tag{2.5}$$

The terminal problem

$$L^{-}_{a,b}p = F \text{ in } Q, \quad p = k \text{ on } \Sigma, \quad p(T,\cdot) = 0 \text{ in } \Omega$$

is also well posed for $F \in \mathbb{Y}$ and $k \in \mathbb{H}_-$. In particular, the map $L^{-}_{a,b} : \mathbb{X}_- \to \mathbb{Y}$ is an isomorphism.

Proof. We begin by deriving a uniform *a priori* estimate for the shifted heat equation $(P^{+}_{b} + \lambda_0)z = g$ for any fixed $\lambda_0 \ge 1$. By [DHP07, Theorem 2.1], this equation admits a unique solution $z \in \mathbb{X}_+$ for any $g \in \mathbb{Y}$. We first derive the energy identities for smooth data. The general case $g \in \mathbb{Y}$ follows by approximation and maximal regularity. Testing $(P^{+}_{b} + \lambda_0)z = g$ with $\bar{z}$, integrating over Q, and taking real parts yield

$$\frac{1}{2}\|z(T)\|^2_{L^2(\Omega)} + \int_Q b|\nabla z|^2 \,\mathrm{d}x\,\mathrm{d}t + \lambda_0\|z\|^2_{\mathbb{Y}} = \mathrm{Re}\int_Q g\bar{z} \,\mathrm{d}x\,\mathrm{d}t.$$

Dropping the nonnegative endpoint term and using $b \ge b_* > 0$ and the Cauchy–Schwarz inequality, we obtain

$$\|z\|_{\mathbb{Y}} \le \lambda_0^{-1}\|g\|_{\mathbb{Y}}, \quad \|\nabla z\|_{\mathbb{Y}} \le b_*^{-1/2}\lambda_0^{-1/2}\|g\|_{\mathbb{Y}}.$$

We next test by $\bar{z}_t$ to obtain

$$\|z_t\|^2_{\mathbb{Y}} + \frac{1}{2}\int_\Omega \Big(b|\nabla z|^2 + \lambda_0|z|^2\,dx\Big)\Big|_0^T = \mathrm{Re}\int_Q g\bar{z}_t \,\mathrm{d}x\,\mathrm{d}t + \frac{1}{2}\int_Q b_t|\nabla z|^2 \,\mathrm{d}x\,\mathrm{d}t.$$

Using $z(0) = 0$ in $H^1(\Omega)$, we can get

$$\|z_t\|^2_{\mathbb{Y}} \le \|g\|_{\mathbb{Y}}\|z_t\|_{\mathbb{Y}} + \frac{1}{2}\|b_t\|_{L^\infty(Q)}\|\nabla z\|^2_{\mathbb{Y}}.$$

By Young's inequality and the estimate for ∇z, we have $\|z_t\|_{\mathbb{Y}} \le C\|g\|_{\mathbb{Y}}$ uniformly for $\lambda_0 \ge 1$. Regarding $-\operatorname{div}(b\nabla z) = g - z_t - \lambda_0 z$ as a stationary elliptic problem for a.e. t, and invoking Dirichlet elliptic regularity, we conclude that

$$\|z\|_{\mathbb{W}} + \lambda_0\|z\|_{\mathbb{Y}} + \sqrt{\lambda_0}\|\nabla z\|_{\mathbb{Y}} \le C\|g\|_{\mathbb{Y}}, \quad \lambda_0 \ge 1, \tag{2.6}$$

where the constant $C > 0$ is independent of λ_0.

We now turn to the inhomogeneous Volterra problem. Using the trace lifting operator, we set $v = \tilde{h} + w$ with $\tilde{h} = E_+h$ and $w \in \mathbb{X}_+$, which reduces the equation (2.4) to

$$P^{+}_{b}w - K_+D_a w = \tilde{F}, \tag{2.7}$$

where $\tilde{F} = F - P^{+}_{b}\tilde{h} + K_+D_a\tilde{h}$ and satisfies

$$\|\tilde{F}\|_{\mathbb{Y}} \le C\Big(\|F\|_{\mathbb{Y}} + \|h\|_{\mathbb{H}}\Big). \tag{2.8}$$

Let $w = e^{\lambda_0 t} z$. We see that

$$(P_b^+ + \lambda_0) z - V_{\lambda_0} D_a z = e^{-\lambda_0 t} \tilde{F}, \tag{2.9}$$

where $(V_{\lambda_0} q)(t) = \int_0^t e^{-\lambda_0 (t-s)} q(s)\, ds$. Let $S_{\lambda_0} = (P_b^+ + \lambda_0)^{-1}$ with homogeneous initial and lateral data. The relevant bounded maps are

$$D_a : \mathbb{X}_+ \longrightarrow \mathbb{Y}, \quad V_{\lambda_0} : \mathbb{Y} \longrightarrow \mathbb{Y}, \quad S_{\lambda_0} : \mathbb{Y} \longrightarrow \mathbb{X}_+.$$

Estimate (2.6) shows that $S_{\lambda_0} : \mathbb{Y} \to \mathbb{X}_+$ is bounded uniformly for $\lambda_0 \geq 1$.

Given $z \in \mathbb{X}_+$, we consider the following problem:

$$\begin{cases} (P_b^+ + \lambda_0)\tilde{z} = V_{\lambda_0} D_a z + e^{-\lambda_0 t} \tilde{F} & \text{in } Q, \\ \tilde{z} = 0, & \text{on } \Sigma \\ \tilde{z}(0) = 0 & \text{in } \Omega. \end{cases} \tag{2.10}$$

Since $\|V_{\lambda_0}\|_{\mathbb{Y} \to \mathbb{Y}} \leq \lambda_0^{-1}$ and $D_a : H^2(\Omega) \to L^2(\Omega)$ is bounded, we have $V_{\lambda_0} D_a z + e^{-\lambda_0 t} \tilde{F} \in \mathbb{Y}$. By maximal regularity for $P_b^+ + \lambda_0$, problem (2.10) has a unique solution $\tilde{z} \in \mathbb{X}_+$. Moreover, estimate (2.6) gives

$$\|\tilde{z}\|_{\mathbb{X}_+} \leq C \|V_{\lambda_0} D_a z + e^{-\lambda_0 t} \tilde{F}\|_{\mathbb{Y}} \leq C C_a \lambda_0^{-1} \|z\|_{\mathbb{X}_+} + C \|\tilde{F}\|_{\mathbb{Y}}. \tag{2.11}$$

Define $\mathfrak{T} : \mathbb{X}_+ \to \mathbb{X}_+$ by $\mathfrak{T} z = \tilde{z}$, where $\tilde{z}$ is the solution of (2.10). For $z_1, z_2 \in \mathbb{X}_+$,

$$\|\mathfrak{T} z_1 - \mathfrak{T} z_2\|_{\mathbb{X}_+} \leq C \|V_{\lambda_0} D_a (z_1 - z_2)\|_{\mathbb{Y}} \leq C C_a \lambda_0^{-1} \|z_1 - z_2\|_{\mathbb{X}_+},$$

where C_a is the operator norm of $D_a : \mathbb{X}_+ \to \mathbb{Y}$. Choose $\lambda_0 \geq 1$ so that $C C_a / \lambda_0 < 1$. Banach's fixed-point theorem then gives a unique $z \in \mathbb{X}_+$ solving (2.9). At the fixed point, estimate (2.11) allows the term involving $\|z\|_{\mathbb{X}_+}$ to be absorbed. Returning to $v = \tilde{h} + e^{\lambda_0 t} z$ and using (2.8) yields (2.5). The terminal result follows by time reversal. □

Proposition 2.2. *Let $h \in \mathbb{H}_+$ and set $f = K_+ h$. Then $v \in \mathbb{W}$ solves (2.4) with $F = 0$ if and only if $u = K_+ v$ solves (1.1) and belongs to $H^2(0,T; L^2(\Omega)) \cap H^1(0,T; H^2(\Omega))$. Thus the displacement-to-traction map induces a bounded boundary-velocity-to-traction map*

$$\mathcal{R}_{a,b}^+ : \mathbb{H}_+ \to \mathbb{N}, \quad h \longmapsto \Lambda_{a,b}^T (K_+ h) = b\, \partial_\nu v + a K_+ \partial_\nu v. \tag{2.12}$$

Proof. If $v \in \mathbb{W}$, then $u = K_+ v \in H^2(0,T; L^2(\Omega)) \cap H^1(0,T; H^2(\Omega))$. Since a is independent of time, $D_a u = K_+ D_a v$. Together with the initial and lateral traces, these identities show that a solution v of (2.4) with $F = 0$ yields a solution u of (1.1).

Hence, if u belongs to the stated space and solves (1.1), then $v = u_t \in \mathbb{W}$ satisfies (2.4) with $F = 0$ and boundary data h. Finally, $\partial_\nu K_+ v = K_+ \partial_\nu v$. As K_+ and multiplication by the smooth boundary coefficients are bounded on $\mathbb{N}$, the estimate (2.5) gives

$$\|\mathcal{R}_{a,b}^+ h\|_{\mathbb{N}} \leq C \|h\|_{\mathbb{H}},$$

which proves that $\mathcal{R}_{a,b}^+ : \mathbb{H}_+ \to \mathbb{N}$ is bounded. □

2.2. **Boundary and volume integral identities.** Define the boundary operators

$$\mathcal{C}_{a,b}^{\pm} z = b\, \partial_\nu z + a K_{\pm} \partial_\nu z.$$

For $k \in \mathbb{H}_-$, let p solve $L_{a,b}^- p = 0$ in Q, with $p = k$ on Σ and $p(T, \cdot) = 0$. We define the auxiliary terminal boundary map by $\mathcal{R}_{a,b}^- k = \mathcal{C}_{a,b}^- p$. The pairings in Green's formula and the inverse identities are complex bilinear. Green's formula gives, for $v, p \in \mathbb{W}$,

$$\int_Q \left((L_{a,b}^+ v) p - v L_{a,b}^- p \right) \mathrm{d}x\, \mathrm{d}t = \int_\Omega v p \, \mathrm{d}x \Big|_0^T + \int_\Sigma \left(v \mathcal{C}_{a,b}^- p - (\mathcal{C}_{a,b}^+ v) p \right) \mathrm{d}S\, \mathrm{d}t. \tag{2.13}$$

For the Volterra terms, we use $K_+^* = K_-$, Fubini's theorem, and the spatial Green formula for D_a. The time independence of a ensures that D_a commutes with $K_\pm$. The identity follows first for smooth functions and extends to $\mathbb{W}$ by approximation and the trace estimates. We have the following two useful integral identities.

Lemma 2.3. *Let $h, k \in C_c^\infty(\Sigma)$, and let $v_1, p_2 \in \mathbb{W}$ be the corresponding solutions of*

$$L^+_{a_1,b_1} v_1 = 0, \quad v_1|_\Sigma = h, \quad v_1(0,\cdot) = 0, \quad L^-_{a_2,b_2} p_2 = 0, \quad p_2|_\Sigma = k, \quad p_2(T,\cdot) = 0.$$

Then, under the condition (1.5),

$$\mathcal{B}(v_1, p_2) := \int_\Sigma \Big((\mathcal{C}^+_{a_1,b_1} v_1) p_2 - v_1 \mathcal{C}^-_{a_2,b_2} p_2 \Big) \, \mathrm{d}S \, \mathrm{d}t = 0 \tag{2.14}$$

and

$$\int_Q (b_1 - b_2) \nabla v_1 \cdot \nabla p_2 \, \mathrm{d}x \, \mathrm{d}t + \int_Q (a_1 - a_2) \nabla (K_+ v_1) \cdot \nabla p_2 \, \mathrm{d}x \, \mathrm{d}t = 0. \tag{2.15}$$

Proof. For either coefficient pair, applying (2.13) to the corresponding forward and terminal solutions gives

$$\int_\Sigma (\mathcal{R}^+_{a_j,b_j} h) k \, \mathrm{d}S \, \mathrm{d}t = \int_\Sigma h (\mathcal{R}^-_{a_j,b_j} k) \, \mathrm{d}S \, \mathrm{d}t, \quad j = 1, 2,$$

because the endpoint term vanishes. Since $K_+ h \in \mathcal{D}_T$, assumption (1.5) implies $\mathcal{R}^+_{a_1,b_1} h = \mathcal{R}^+_{a_2,b_2} h$. The transposition identity for the second coefficient pair therefore yields

$$\mathcal{B}(v_1, p_2) = \int_\Sigma \Big((\mathcal{R}^+_{a_1,b_1} - \mathcal{R}^+_{a_2,b_2}) h \Big) k \, \mathrm{d}S \, \mathrm{d}t = 0.$$

Next, we apply (2.13) for (a_1, b_1) to v_1 and p_2, and use $L^-_{a_2,b_2} p_2 = 0$. Integration by parts and $K_+^* = K_-$ give

$$\mathcal{B}(v_1, p_2) = \int_Q (b_1 - b_2) \nabla v_1 \cdot \nabla p_2 \, \mathrm{d}x \, \mathrm{d}t + \int_Q (a_1 - a_2) \nabla (K_+ v_1) \cdot \nabla p_2 \, \mathrm{d}x \, \mathrm{d}t.$$

Here the time independence of $a_1 - a_2$ allows it to pass through $K_\pm$. Combining this equality with $\mathcal{B}(v_1, p_2) = 0$ proves (2.15), as desired. □

Remark 2.4. We will use the boundary identity (2.14) to recover b without prior knowledge of a in the interior. Once b is known, the volume identity (2.15) yields the recovery of a. The proof uses only measurements with boundary inputs $f = K_+ h$, $h \in C_c^\infty(\Sigma)$, and hence requires $\Lambda^T_{a,b}$ only on the class $\mathcal{D}_T$ specified in Theorem 1.1.

3. Boundary determination of the coefficients

In this section, we determine all normal derivatives of the coefficients a and b on the boundary from the displacement-to-traction map through the construction of boundary-localized, highly oscillatory solutions. Throughout this section, all equations and measurement maps are complexified, and pairings of complex spatial covectors are bilinear.

We now state the following boundary determination result for a and b.

Proposition 3.1. *Let (a_j, b_j), $j = 1, 2$, satisfy the conditions of Theorem 1.1, and suppose that (1.5) holds. Then, for every $m \geq 0$,*

$$\partial_\nu^m a_1 = \partial_\nu^m a_2 \quad \text{on } \Gamma, \quad \partial_\nu^m b_1 = \partial_\nu^m b_2 \quad \text{on } [0, T] \times \Gamma. \tag{3.1}$$

3.1. **Construction of the phase and amplitude functions.** Fix $(t_0, x_0) \in (0,T) \times \Gamma$, and let U be a sufficiently small neighborhood of x_0. We choose Euclidean boundary normal coordinates $y = (y_1, y')$ on U such that $y_1 = 0$ on $\Gamma \cap U$ and $y_1 < 0$ in $\Omega \cap U$. Let U' denote the y'-coordinate image of $\Gamma \cap U$. In these coordinates,

$$\nu = \partial_{y_1} \quad \text{on } \Gamma \cap U, \quad e = \mathrm{d}y_1^2 + e_{\alpha\beta}(y)\, \mathrm{d}y_\alpha\, \mathrm{d}y_\beta.$$

Thus $\rho = -y_1 > 0$ is the inward normal distance in $\Omega \cap U$.

We next construct an approximate geometric optics solution localized near the boundary, taking the following WKB form

$$v_\tau^{\mathrm{app}} = e^{i\tau t + \tau\Phi} \vartheta w_\tau^{\mathrm{amp}}, \quad w_\tau^{\mathrm{amp}} = \sum_{k=0}^{N-1} \tau^{-k} w_k, \tag{3.2}$$

where $\tau > 0$ is a large parameter, $\Phi(y)$ is a time-independent complex phase function, and $\vartheta \in C_c^\infty(\mathbb{R})$ is a cutoff function in the normal variable y_1, equal to 1 for $|y_1| \le \varepsilon/4$ and vanishing for $|y_1| \ge \varepsilon/2$. The amplitudes w_k satisfy the same boundary condition $w_k|_\Gamma = \delta_{k0}\chi$, where $\chi \in C_c^\infty((0,T) \times U')$.

Direct calculation of the conjugated operator $\mathcal{P}_\tau w = e^{-i\tau t - \tau\Phi} L_{a,b}^+ (e^{i\tau t + \tau\Phi} w)$ yields

$$\mathcal{P}_\tau w = -\tau^2 b E_\Phi w - \tau T_b w + P_b^+ w - \mathcal{K}_\tau(\tau^2 a E_\Phi w + \tau S_a w + D_a w), \tag{3.3}$$

where the eikonal defect is $E_\Phi := \langle d\Phi, d\Phi\rangle_e$, and the transport and memory operators are defined as

$$T_b w = 2b\langle d\Phi, dw\rangle_e + (\langle db, d\Phi\rangle_e + b\Delta\Phi - i)w, \tag{3.4}$$

$$S_a w = 2a\langle d\Phi, dw\rangle_e + (\langle da, d\Phi\rangle_e + a\Delta\Phi)w, \tag{3.5}$$

$$\mathcal{K}_\tau F = e^{-i\tau t} K_+ (e^{i\tau t} F). \tag{3.6}$$

To eliminate the terms of order τ^2 in (3.3), we construct the phase function Φ of the following form

$$\Phi(y) = \sum_{j=0}^{N} \frac{y_1^j}{j!} \Phi_j(y').$$

We require E_Φ to vanish up to order $N-1$ at the boundary $y_1 = 0$. For $j = 0$, we have

$$\Phi_1^2 + e^{\alpha\beta}(0, y')\partial_\alpha \Phi_0 \partial_\beta \Phi_0 = 0.$$

Choose a real-valued smooth function ψ on U' satisfying $|d\psi|^2_{e|_\Gamma} = 1$, and set $\Phi_0 = i\psi$ and $\Phi_1 = 1$. Then $E_\Phi|_\Gamma = 0$. For $1 \le j \le N-1$, the condition $\partial_{y_1}^j E_\Phi|_\Gamma = 0$ uniquely determines Φ_{j+1} from $\Phi_0, \cdots, \Phi_j$. Taylor's theorem therefore gives $E_\Phi = O(\rho^N)$. Moreover, since $\operatorname{Re}\Phi = -\rho + O(\rho^2)$, we have $\operatorname{Re}\Phi \le -c\rho$ for $0 \le \rho < \varepsilon$, provided $\varepsilon > 0$ is sufficiently small.

Let X be a Hilbert space, and let F be a smooth X-valued function that vanishes near $t = 0$. Repeated integration by parts in time yields the following expansion for $\mathcal{K}_\tau F$:

$$\mathcal{K}_\tau F = \sum_{r=0}^{N_{\mathrm{time}}-1} \tau^{-r-1} c_r \partial_t^r F + \mathcal{E}_{N_{\mathrm{time}},\tau} F, \tag{3.7}$$

where $c_r = (-1)^r i^{-r-1}$ and

$$\mathcal{E}_{N_{\mathrm{time}},\tau} F = (-1)^{N_{\mathrm{time}}} (i\tau)^{-N_{\mathrm{time}}} \mathcal{K}_\tau(\partial_t^{N_{\mathrm{time}}} F).$$

A further integration by parts, together with the boundedness of K_+ on $L^2(0,T;X)$, yields the remainder estimate

$$\|\mathcal{E}_{N_{\mathrm{time}},\tau} F\|_{L^2(0,T;X)} \le C_T \tau^{-N_{\mathrm{time}}-1} \Big(\|\partial_t^{N_{\mathrm{time}}} F\|_{L^2(0,T;X)} + \|\partial_t^{N_{\mathrm{time}}+1} F\|_{L^2(0,T;X)} \Big). \tag{3.8}$$

Substituting the expansion into the memory terms of (3.3), we obtain the contribution at order τ^{1-k} for $k \geq 1$

$$M_k = -\sum_{r+q=k-1} c_r \partial_t^r S_a w_q - \sum_{r+q=k-2} c_r \partial_t^r D_a w_q. \tag{3.9}$$

Since $E_\Phi = O(\rho^N)$, the terms containing E_Φ in (3.3) have vanishing normal derivatives on Γ through order $N-1$. Matching the remaining powers of τ gives the following relations for the normal derivatives of the amplitudes at the boundary:

$$\partial_\nu^j (T_b w_0)|_\Gamma = 0, \qquad 0 \leq j \leq N-1, \tag{3.10}$$

$$\partial_\nu^j (-T_b w_k + P_b^+ w_{k-1} + M_k)|_\Gamma = 0, \qquad 1 \leq k \leq N-1, \quad 0 \leq j \leq N-k-1. \tag{3.11}$$

For $m \geq 0$, let $J_\nu^m c = (\partial_\nu^j c|_\Sigma)_{j=0}^m$ denote the collection of normal derivatives of c up to order m, and let $J_\nu^{-1} c$ denote the empty tuple. Set $\beta = b|_\Sigma$ and $\alpha = a|_\Gamma$. For $0 \leq k \leq N-1$ and $0 \leq \ell \leq N-k$, set $W_{k,\ell} = (\partial_\nu^\ell w_k)|_\Sigma$.

The following lemma describes how the boundary equations determine the normal derivatives of the amplitudes in terms of those of a and b.

Lemma 3.2. *For $1 \leq \ell \leq N$,*

$$W_{0,\ell} = -\frac{\chi}{2\beta}\, \partial_\nu^\ell b + F_{0,\ell}(J_\nu^{\ell-1} b). \tag{3.12}$$

For $k, \ell \geq 1$ with $k + \ell \leq N$,

$$W_{k,\ell} = \left(-\frac{1}{2}\right)^{k+1} \frac{\chi}{\beta}\, \partial_\nu^{k+\ell} b + \left(-\frac{1}{2}\right)^{k} \frac{\chi}{i\beta}\, \partial_\nu^{k+\ell-1} a + F_{k,\ell}(J_\nu^{k+\ell-1} b, J_\nu^{k+\ell-2} a). \tag{3.13}$$

Here, $F_{k,\ell}$ are smooth differential expressions depending strictly on the indicated lower-order normal derivatives of a, b, their tangential and time derivatives, and the known functions χ, e, Φ and β.

Proof. We argue by induction on $s = k + \ell$, determining

$$W_{0,s},\ W_{1,s-1},\ \ldots,\ W_{s-1,1}$$

in this order. The values $W_{k,0} = \delta_{k0}\chi$ are prescribed. For $k = 0$, the equation with $j = \ell - 1$ in (3.10) has the leading terms $2\beta W_{0,\ell} + (\partial_\nu^\ell b)\chi$. All other terms involve lower-order normal derivatives of b. Since $\beta > 0$, this determines $W_{0,\ell}$ and gives (3.12).

For $k \geq 1$, the equation with $j = \ell - 1$ in (3.11) contains the terms

$$-2\beta W_{k,\ell} - \beta W_{k-1,\ell+1}. \tag{3.14}$$

The second amplitude derivative is already known at the current induction step. In the two sums defining M_k, the indices satisfy $q \leq k-1$ and $q \leq k-2$, respectively. Counting the normal derivatives in $S_a w_q$ and $D_a w_q$ shows that these sums involve normal derivatives of b of order at most $s-1$. Thus the derivative $\partial_\nu^s b$ enters through $W_{k-1,\ell+1}$, and its coefficient is multiplied by $-1/2$ at each step.

When $k = 1$, the term $-i^{-1} S_a w_0$ in M_1 contributes $-i^{-1}(\partial_\nu^\ell a)\chi$. Solving (3.14) therefore gives the coefficient $-(2i\beta)^{-1}$ of $\partial_\nu^\ell a$. For $k \geq 2$, the direct terms involve normal derivatives of a of order at most $s-2$. Hence the derivative $\partial_\nu^{s-1} a$ also propagates through $W_{k-1,\ell+1}$ with a factor of $-1/2$ at each step. This establishes (3.13). Since $\beta > 0$, the recursion uniquely determines all $W_{k,\ell}$. □

Using the determined boundary derivatives $W_{k,\ell}$, we construct the polynomial amplitudes of the following form

$$w_k(t,y) = \sum_{\ell=0}^{N-k} \frac{y_1^\ell}{\ell!} W_{k,\ell}(t,y'). \tag{3.15}$$

Equations (3.12) and (3.13) show that each $W_{k,\ell}$ is smooth and supported in $\operatorname{supp}\chi$. Hence its extension by zero outside U' is smooth.

3.2. **Remainder estimates and proof of Proposition 3.1.** Let the remainder r_τ satisfy the following problem

$$\begin{cases} L_{a,b}^+ r_\tau = F_\tau & \text{in } Q, \\ r_\tau = 0 & \text{on } \Sigma, \\ r_\tau|_{t=0} = 0 & \text{in } \Omega, \end{cases}$$

with source $F_\tau = -L_{a,b}^+ v_\tau^{\mathrm{app}}$. Set $w_{-1} = w_N = 0$ and $M_0 = 0$. Expanding $F_\tau = -L_{a,b}^+ v_\tau^{\mathrm{app}}$ by (3.3) and (3.7) gives

$$F_\tau = e^{i\tau t+\tau\Phi}\vartheta\Big(\tau^2 bE_\Phi w_\tau^{\mathrm{amp}} + \tau^2\mathcal{K}_\tau(aE_\Phi w_\tau^{\mathrm{amp}}) + \sum_{k=0}^{N}\tau^{1-k}\mathcal{F}_k + \mathcal{T}_{N,N_{\mathrm{time}},\tau} + \widetilde{\mathcal{E}}_{N_{\mathrm{time}},\tau}\Big) + \mathcal{C}_{\vartheta,\tau},$$

where $\mathcal{F}_k = T_b w_k - P_b^+ w_{k-1} - M_k$, $\mathcal{T}_{N,N_{\mathrm{time}},\tau}$ collects the remaining algebraic terms from the finite temporal expansion, and

$$\widetilde{\mathcal{E}}_{N_{\mathrm{time}},\tau} = \sum_{k=0}^{N-1}\left(\tau^{1-k}\mathcal{E}_{N_{\mathrm{time}},\tau}(S_a w_k) + \tau^{-k}\mathcal{E}_{N_{\mathrm{time}},\tau}(D_a w_k)\right).$$

Every term in $\mathcal{T}_{N,N_{\mathrm{time}},\tau}$ has a power at most τ^{-N}. The cutoff commutator $\mathcal{C}_{\vartheta,\tau}$ is supported where $\nabla\vartheta \neq 0$ and is exponentially small.

We use the normal decay estimate

$$\|\rho^q e^{-c\tau\rho}\|_{L^2(0,\varepsilon)} \le C_q \tau^{-q-1/2}, \quad q \ge 0. \tag{3.16}$$

Since $E_\Phi = O(\rho^N)$ and the amplitudes are independent of τ, (3.16) gives

$$\|e^{\tau\Phi}\vartheta\,\tau^2 bE_\Phi w_\tau^{\mathrm{amp}}\|_{L^2(Q)} = O(\tau^{3/2-N}).$$

The amplitudes vanish near $t = 0$. As aE_Φ is independent of time, integration by parts gives the identity

$$\tau^2\mathcal{K}_\tau(aE_\Phi w_\tau^{\mathrm{amp}}) = -i\tau aE_\Phi w_\tau^{\mathrm{amp}} + i\tau\mathcal{K}_\tau(aE_\Phi\partial_t w_\tau^{\mathrm{amp}}).$$

The bound for $\mathcal{K}_\tau$ on $L^2(Q)$ and (3.16) therefore imply

$$\left\|e^{\tau\Phi}\vartheta\tau^2\mathcal{K}_\tau(aE_\Phi w_\tau^{\mathrm{amp}})\right\|_{L^2(Q)} = O(\tau^{1/2-N}).$$

Equations (3.10) and (3.11) give $\mathcal{F}_k = O(\rho^{N-k})$ for $0 \le k < N$. Moreover, the remaining coefficient $\mathcal{F}_N$ is bounded. Hence, for $0 \le k \le N$, we have

$$\left\|e^{\tau\Phi}\vartheta\tau^{1-k}\mathcal{F}_k\right\|_{L^2(Q)} = O(\tau^{1/2-N}).$$

The estimate (3.8) bounds the two temporal remainders for $S_a w_k$ and $D_a w_k$ by $O(\tau^{-N_{\mathrm{time}}-1})$. Since the time-independent weight $\omega_\tau = e^{\tau\Phi}\vartheta$ commutes with the remainder operator, we apply (3.8) to $\omega_\tau G$, with $G = S_a w_k$ or $G = D_a w_k$, and use (3.16) and the smoothness of G to bound the required time derivatives of $\omega_\tau G$ by $O(\tau^{-1/2})$ in $L^2(Q)$. Thus

$$\|e^{\tau\Phi}\vartheta\tau^{1-k}\mathcal{E}_{N_{\mathrm{time}},\tau}(S_a w_k)\|_{L^2(Q)} = O(\tau^{-N_{\mathrm{time}}-k-1/2}),$$
$$\|e^{\tau\Phi}\vartheta\tau^{-k}\mathcal{E}_{N_{\mathrm{time}},\tau}(D_a w_k)\|_{L^2(Q)} = O(\tau^{-N_{\mathrm{time}}-k-3/2}).$$

Finally, since $\operatorname{supp}(\nabla\vartheta)$ is disjoint from the boundary Γ, the commutator $\mathcal{C}_{\vartheta,\tau}$ decays as $O(\tau^{-L})$ for any fixed $L > 0$.

Combining these estimates and choosing $N \geq M+4$ and $N_{\mathrm{time}} \geq N+3$ gives

$$\|F_\tau\|_{L^2(Q)} \leq C_{N,N_{\mathrm{time}}}\left(\tau^{3/2-N} + \tau^{1/2-N} + \tau^{-N_{\mathrm{time}}-1/2} + \tau^{-N_{\mathrm{time}}-3/2}\right) \leq C_M \tau^{-M-2}. \quad (3.17)$$

By Proposition 2.1, we obtain the estimate

$$\|r_\tau\|_{\mathbb{W}} + \|\partial_\nu r_\tau\|_{\mathbb{N}} \leq C_M \tau^{-M-2}. \quad (3.18)$$

Thus $V_\tau = v_\tau^{\mathrm{app}} + r_\tau$ solves (2.1). Its boundary velocity is $h_\tau(t,y') = e^{i\tau(t+\psi(y'))}\chi(t,y')$. Let $\Sigma' := (0,T)\times U'$ denote the local lateral boundary in the coordinates y'. The phase ψ is defined on U', and χ is compactly supported in Σ'. Since $\partial_\nu \Phi|_\Gamma = 1$ and the cutoff is identically 1 near the boundary, the expansion

$$e^{-i\tau(t+\psi)}\partial_\nu V_\tau = \tau\chi + \sum_{k=0}^{M} \tau^{-k} W_{k,1} + O_{L^2(\Sigma')}(\tau^{-M-1})$$

holds on Σ'. Removing the phase in (2.12) turns K_+ into $\mathcal{K}_\tau$, since ψ is independent of time. Applying (3.7) to the smooth terms and using the L^2 boundedness of $\mathcal{K}_\tau$ for the trace remainder gives

$$e^{-i\tau(t+\psi)}\mathcal{R}^+_{a,b} h_\tau = \tau\beta\chi + \sum_{m=0}^{M} \tau^{-m} B_m + O_{L^2(\Sigma')}(\tau^{-M-1}), \quad (3.19)$$

where

$$B_m = \beta W_{m,1} + \alpha c_m \partial_t^m \chi + \alpha \sum_{\substack{r,k\geq 0 \\ r+k=m-1}} c_r \partial_t^r W_{k,1}, \qquad c_r = (-1)^r i^{-r-1}.$$

The sum is empty for $m = 0$. Substituting (3.12) and (3.13) into this formula yields

$$B_m = \left(-\frac{1}{2}\right)^{m+1} \partial_\nu^{m+1} b\,\chi + \left(-\frac{1}{2}\right)^{m} i^{-1}\partial_\nu^m a\,\chi + H_m(J_\nu^m b, J_\nu^{m-1} a). \quad (3.20)$$

Here H_m depends only on normal derivatives of b through order m and of a through order $m-1$, together with their tangential derivatives, the relevant time derivatives, and the fixed phase and boundary profile. In particular, the first two orders are given by

$$B_0 = -\tfrac{1}{2}\partial_\nu b\,\chi - i\alpha\chi + H_0, \quad (3.21)$$

$$H_0 = -i\beta\langle d_\Gamma\psi, d_\Gamma\chi\rangle - \tfrac{i}{2}\langle d_\Gamma\beta, d_\Gamma\psi\rangle\chi - \tfrac{\beta}{2}(\Delta\Phi)|_\Gamma\chi + \tfrac{i}{2}\chi,$$

$$B_1 = \tfrac{1}{4}\partial_\nu^2 b\,\chi + \tfrac{i}{2}\partial_\nu a\,\chi + H_1(J_\nu^1 b, J_\nu^0 a). \quad (3.22)$$

We are now in a position to prove Proposition 3.1.

Proof of Proposition 3.1. For $j = 1,2$, we apply (3.19) to the response $\mathcal{R}^+_{a_j,b_j} h_\tau$. Since $f_\tau = K_+ h_\tau \in \mathcal{D}_T$, the condition (1.5) gives $\mathcal{R}^+_{a_1,b_1} h_\tau = \mathcal{R}^+_{a_2,b_2} h_\tau$. Dividing the difference of the expansions by τ and letting $\tau\to\infty$ yields

$$(b_1 - b_2)|_\Sigma\,\chi = 0.$$

Using the arbitrariness of the real-valued function χ, we have $b_1 = b_2$ on Σ.

We argue by induction. Suppose that $J_\nu^m b_1 = J_\nu^m b_2$ and $J_\nu^{m-1} a_1 = J_\nu^{m-1} a_2$ on Σ. When we subtract the two expansions through order τ^{-m}, the coefficients of all powers higher than τ^{-m} and both H_m terms cancel. Multiplying by τ^m and passing to the limit in $L^2(\Sigma')$ gives

$$\left(-\frac{1}{2}\right)^{m+1} \partial_\nu^{m+1}(b_1 - b_2)\,\chi + \left(-\frac{1}{2}\right)^{m} i^{-1}\partial_\nu^m (a_1 - a_2)\,\chi = 0.$$

The coefficients and χ are real-valued, so the real and imaginary parts give $\partial_\nu^{m+1}(b_1-b_2)\chi = 0$ and $\partial_\nu^m(a_1-a_2)\chi = 0$, respectively. Since both the local coordinate neighborhood on Γ and the test function χ are arbitrary, we obtain $\partial_\nu^{m+1}b_1 = \partial_\nu^{m+1}b_2$ and $\partial_\nu^m a_1 = \partial_\nu^m a_2$ on Σ. This completes the induction.

Differentiating these equalities along Γ and, for the time-dependent coefficients b_j, in time gives the corresponding identities for tangential and time derivatives. By smoothness, the identities for b_1 and b_2, including those for their derivatives, extend to $t = 0$ and $t = T$. Thus, the proof is finished. □

4. Construction of exponential solutions with memory

By Proposition 3.1, $b_2 - b_1$ vanishes to infinite order on $[0,T]\times\Gamma$. Extending b_1 smoothly and $b_2 - b_1$ by zero gives smooth extensions $\widetilde{b}_1$ and $\widetilde{b}_2$ to a slightly larger smooth domain $\widetilde{\Omega}$, with

$$\widetilde{b}_1 = \widetilde{b}_2 \quad \text{on } [0,T]\times(\widetilde{\Omega}\setminus\Omega).$$

For a sufficiently small enlargement, both extensions are positive. By openness of simplicity and compactness of $[0,T]$, the extended metrics $\widetilde{b}_j^{-1}(t,\cdot)e$ remain simple for every $t\in[0,T]$.

For each $t\in[0,T]$, write

$$g_{b_j}(t) = b_j^{-1}(t,\cdot)e, \qquad \widetilde{g}_j(t) = \widetilde{b}_j^{-1}(t,\cdot)e \tag{4.1}$$

for the metrics on $\overline{\Omega}$ and $\overline{\widetilde{\Omega}}$, respectively. We denote the distance between $x,y\in\overline{\widetilde{\Omega}}$ for $\widetilde{g}_j(t)$ by $d_j^e(t;x,y)$, and the intrinsic distance between $p,q\in\overline{\Omega}$ for $g_{b_j}(t)$ by $d_j^\Omega(t;p,q)$.

In this section, we aim to construct high-frequency exponential solutions for the following forward and backward equations,

$$\begin{cases} L_{a,b}^+ v_\tau = 0 \\ v_\tau(0) = 0 \end{cases} \quad \text{and} \quad \begin{cases} L_{a,b}^- p_\tau = 0 \\ p_\tau(T) = 0 \end{cases}$$

of the form

$$v_\tau = e^{\Theta_\tau}(\mu_+ + R_{+,\tau}), \quad p_\tau = e^{-\Theta_\tau}(\mu_- + R_{-,\tau}),$$

where τ is a large parameter, $\Theta_\tau = \tau^2 t + \tau\phi(t,x)$, and $R_{\pm,\tau}$ denote the remainder terms that tend to zero as $\tau\to\infty$ in suitable norms. The conjugated time kernel has L^1 norm $O(\tau^{-2})$, which compensates for the two spatial derivatives in the memory term.

4.1. WKB expansion for phase and amplitudes.

Substituting $e^{\pm\Theta_\tau}\mu_\pm$ into $P_b^\pm$, we have

$$e^{\mp\Theta_\tau}P_b^\pm\big(e^{\pm\Theta_\tau}\mu_\pm\big) = \tau^2\big(1-b|\nabla\phi|^2\big)\mu_\pm \mp \tau\mathcal{T}_\phi^\pm\mu_\pm + P_b^\pm\mu_\pm, \tag{4.2}$$

where the operator $\mathcal{T}_\phi^\pm$ is defined by

$$\mathcal{T}_\phi^\pm w = 2b\nabla\phi\cdot\nabla w + \big(\operatorname{div}(b\nabla\phi) \mp \phi_t\big)w.$$

To make the leading-order term in (4.2) vanish, we are required to solve the eikonal equation

$$b|\nabla\phi|^2 = 1 \quad \text{on } [0,T]\times\overline{\Omega}. \tag{4.3}$$

For a single viscosity b, write $\widetilde{b}$ for its extension and $\widetilde{g}(t) = \widetilde{b}^{-1}(t,\cdot)e$. For $x_0\in\partial\widetilde{\Omega}$, the distance phase function $\phi(t,x) = d_{\widetilde{g}(t)}(x_0,x)$ solves (4.3). It follows that the amplitudes $\mu_\pm$ satisfy the transport equations

$$\mathcal{T}_\phi^\pm\mu_\pm = 0. \tag{4.4}$$

For an inward-pointing Euclidean unit vector ω at x_0, define the $\widetilde{g}(t)$-geodesic

$$\gamma_{t,x_0,\omega}(r) = \exp_{x_0}^{\widetilde{g}(t)}\Big(r\sqrt{\widetilde{b}(t,x_0)}\,\omega\Big),$$

parametrized by arc length r. In these coordinates, we write $f(t,r,\omega) = f(t,\gamma_{t,x_0,\omega}(r))$. In particular, ϕ_t denotes the time derivative at fixed x, evaluated at $x = \gamma_{t,x_0,\omega}(r)$. For the extended coefficient $\widetilde{b}$, the transport equation (4.4) takes the form

$$2\partial_r\mu_\pm + \Big(\operatorname{div}(\widetilde{b}\nabla\phi) \mp \phi_t\Big)\mu_\pm = 0.$$

Let V have compact closure in the set of inward directions. Choose $r_* > 0$ small enough that $\gamma_{t,x_0,\omega}(r) \in \widetilde{\Omega}\setminus\overline{\Omega}$ for $0 < r \le r_*$, $t \in [0,T]$, and $\omega \in V$. Choose $\chi_\pm \in C_c^\infty((0,T)\times V)$ independently and prescribe $\mu_\pm(t,r_*,\omega) = \chi_\pm(t,\omega)$. Solving the transport equations in r gives

$$\mu_\pm(t,r,\omega) = \chi_\pm(t,\omega)\exp\!\Big(-\frac{1}{2}\int_{r_*}^{r}\Big(\operatorname{div}(\widetilde{b}\nabla\phi) \mp \phi_t\Big)(t,s,\omega)\,\mathrm{d}s\Big).$$

The compact angular support of $\chi_\pm$ allows us to extend $\mu_\pm$ smoothly by zero to directions outside V.

4.2. **Remainder estimates.** Once the phase ϕ and leading amplitudes $\mu_\pm$ are constructed, the remainders $R_{\pm,\tau} \in \mathbb{X}_\pm$ have to satisfy

$$A_\tau^\pm R_{\pm,\tau} = -F_{\pm,\tau}, \tag{4.5}$$

where

$$A_\tau^\pm = e^{\mp\Theta_\tau}L_{a,b}^\pm e^{\pm\Theta_\tau}, \quad F_{\pm,\tau} = P_b^\pm\mu_\pm - \mathcal{M}_\tau^\pm\mu_\pm,$$

with the conjugated memory operators

$$\mathcal{M}_\tau^+ r(t) = e^{-\Theta_\tau(t)}\int_0^t D_a\Big(e^{\Theta_\tau(s)}r(s)\Big)\,\mathrm{d}s, \quad \mathcal{M}_\tau^- r(t) = e^{\Theta_\tau(t)}\int_t^T D_a\Big(e^{-\Theta_\tau(s)}r(s)\Big)\,\mathrm{d}s.$$

To invert $A_\tau^\pm$, we first study the conjugated parabolic operators

$$B_\tau^\pm = e^{\mp\Theta_\tau}P_b^\pm e^{\pm\Theta_\tau} = P_b^\pm \mp \tau\mathcal{T}_\phi^\pm. \tag{4.6}$$

We then treat the Volterra term in $A_\tau^\pm = B_\tau^\pm - \mathcal{M}_\tau^\pm$ as a perturbation. We now state the following weighted energy estimate.

Lemma 4.1. *Let ϕ be the constructed phase function, satisfying (4.3). For each fixed τ, the operators $B_\tau^\pm\colon \mathbb{X}_\pm \to \mathbb{Y}$ are isomorphisms. There exist constants $C > 0$ and $\tau_0 > 0$ such that, for every $\tau \ge \tau_0$ and $F \in \mathbb{Y}$, the unique solution $r \in \mathbb{X}_\pm$ of $B_\tau^\pm r = F$ satisfies*

$$\tau\|r\|_{\mathbb{Y}} + \tau^{1/2}\|\nabla r\|_{\mathbb{Y}} + \tau^{-1/2}\Big(\|r\|_{\mathbb{W}} + \|\partial_\nu r\|_{\mathbb{N}}\Big) \le C\|F\|_{\mathbb{Y}}. \tag{4.7}$$

Proof. Since $P_b^\pm : \mathbb{X}_\pm \to \mathbb{Y}$ are isomorphisms and the weights $e^{\pm\Theta_\tau}$ have bounded inverses, $B_\tau^\pm$ are isomorphisms for each fixed τ.

We consider first the forward operator B_τ^+, and rewrite the equation $B_\tau^+ r = F$ as

$$\partial_t r - \operatorname{div}(b\nabla r) - \tau\mathcal{T}_\phi^+ r = F.$$

Take $\lambda > \|\phi_t\|_\infty + 1$ and set $w = e^{\lambda\phi}$. Testing the equation by $w\bar{r}$, taking the real part, and integrating by parts, we apply $r|_\Sigma = 0$ and (4.3) to obtain

$$\operatorname{Re}\int_Q w\bar{r}\,\mathcal{T}_\phi^+ r\,\mathrm{d}x\,\mathrm{d}t = -\int_Q(\lambda+\phi_t)w|r|^2\,\mathrm{d}x\,\mathrm{d}t.$$

$$\operatorname{Re}\int_Q w\bar{r}\,r_t = \frac{1}{2}\int_\Omega w(T)|r(T)|^2\,\mathrm{d}x - \frac{1}{2}\int_Q w_t|r|^2\,\mathrm{d}x\,\mathrm{d}t,$$

$$\operatorname{Re}\int_Q w\bar{r}(-\operatorname{div}(b\nabla r)) = \int_Q wb|\nabla r|^2\,\mathrm{d}x\,\mathrm{d}t - \frac{1}{2}\int_Q \operatorname{div}(b\nabla w)|r|^2\,\mathrm{d}x\,\mathrm{d}t.$$

We sum these identities and drop the nonnegative endpoint term. For sufficiently large τ, we absorb the lower-order terms, which are bounded by $C\|r\|_{\mathbb{Y}}^2$, into the positive term $c\tau\|r\|_{\mathbb{Y}}^2$. Applying the Cauchy-Schwarz inequality to the source term then yields

$$c\tau\|r\|_{\mathbb{Y}}^2 + c\|\nabla r\|_{\mathbb{Y}}^2 \le C\|F\|_{\mathbb{Y}}\|r\|_{\mathbb{Y}}.$$

Hence, we can get

$$\|r\|_{\mathbb{Y}} \le C\tau^{-1}\|F\|_{\mathbb{Y}}, \quad \|\nabla r\|_{\mathbb{Y}} \le C\tau^{-1/2}\|F\|_{\mathbb{Y}}. \tag{4.8}$$

Since $\mathcal{T}_\phi^+$ is a first-order operator with bounded coefficients, estimate (4.8) gives

$$\tau\|\mathcal{T}_\phi^+ r\|_{\mathbb{Y}} \le C\tau\Big(\|\nabla r\|_{\mathbb{Y}} + \|r\|_{\mathbb{Y}}\Big) \le C\tau^{1/2}\|F\|_{\mathbb{Y}}.$$

Since $P_b^+ r = F + \tau\mathcal{T}_\phi^+ r$, the triangle inequality yields $\|P_b^+ r\|_{\mathbb{Y}} \le C\tau^{1/2}\|F\|_{\mathbb{Y}}$. Applying the estimate (2.6) with $\lambda_0 = 1$, we have

$$\|r\|_{\mathbb{W}} \le C\|(P_b^+ + 1)r\|_{\mathbb{Y}} \le C\Big(\|P_b^+ r\|_{\mathbb{Y}} + \|r\|_{\mathbb{Y}}\Big) \le C\tau^{1/2}\|F\|_{\mathbb{Y}}. \tag{4.9}$$

Combining (4.8), (4.9), and the estimate (2.3) for normal trace yields (4.7).

For the backward problem, we test $B_\tau^- r = F$ against $w\bar{r}$, with $w = e^{-\lambda\phi}$. Integration by parts in space gives

$$\operatorname{Re}\int_Q w\bar{r}\,\mathcal{T}_\phi^- r\;\mathrm{d}x\;\mathrm{d}t = \int_Q (\lambda + \phi_t)w|r|^2\;\mathrm{d}x\,\mathrm{d}t.$$

Since $r(T,\cdot) = 0$, integration by parts in time leaves only a nonnegative endpoint term at $t = 0$. The remaining estimates proceed as in the forward case and yield (4.7). □

Let $G_\tau^\pm$ denote the inverses of $B_\tau^\pm$. In the conjugation of $D_a = \operatorname{div}(a\nabla)$ by $e^{\pm\Theta_\tau}$, the zeroth-, first-, and second-order coefficients have sizes $O(\tau^2)$, $O(\tau)$, and $O(1)$, respectively. To estimate $\mathcal{M}_\tau^\pm$, we therefore equip $\mathbb{Z}_\tau = L^2(0,T;H^2(\Omega))$ with the scaled norm

$$\|r\|_{\mathbb{Z}_\tau} = \|r\|_{\mathbb{Y}} + \tau^{-1}\|\nabla r\|_{\mathbb{Y}} + \tau^{-2}\|D_x^2 r\|_{\mathbb{Y}}.$$

Lemma 4.2. *There exist constants $C > 0$ and $\tau_0 > 0$ such that, for all $\tau \ge \tau_0$,*

$$\|\mathcal{M}_\tau^\pm\|_{\mathbb{Z}_\tau\to\mathbb{Y}} \le C.$$

Moreover,

$$\|\mathcal{M}_\tau^\pm G_\tau^\pm\|_{\mathbb{Y}\to\mathbb{Y}} \le C\tau^{-1}.$$

Proof. Let $L = \|\phi_t\|_\infty$. For $s \le t$ and $\tau \ge 2L + 1$, the forward Volterra kernel satisfies

$$e^{-\Theta_\tau(t,x)+\Theta_\tau(s,x)} \le e^{-\tau^2(t-s)/2},$$

which yields an L^1-norm bound of $2\tau^{-2}$. At each fixed time, conjugation of D_a gives

$$e^{-\Theta_\tau(s)} D_a\Big(e^{\Theta_\tau(s)} r\Big) = D_a r + \tau\Big(2a\nabla\phi\cdot\nabla r + \operatorname{div}(a\nabla\phi)\, r\Big) + \tau^2 a|\nabla\phi|^2 r.$$

Combining the conjugation formula with the kernel bound and applying Young's convolution inequality in time, we obtain

$$\|\mathcal{M}_\tau^+ r\|_{\mathbb{Y}} \le C\tau^{-2}\Big(\tau^2\|r\|_{\mathbb{Y}} + \tau\|\nabla r\|_{\mathbb{Y}} + \|D_a r\|_{\mathbb{Y}}\Big) \le C\|r\|_{\mathbb{Z}_\tau}.$$

The estimate for $\mathcal{M}_\tau^-$ follows by the same argument for $s \ge t$.

To estimate $\mathcal{M}_\tau^\pm G_\tau^\pm$, we fix $F \in \mathbb{Y}$ and set $r = G_\tau^\pm F$. Applying (4.7) to $B_\tau^\pm r = F$ gives

$$\|r\|_{\mathbb{Y}} \le C\tau^{-1}\|F\|_{\mathbb{Y}}, \quad \|\nabla r\|_{\mathbb{Y}} \le C\tau^{-1/2}\|F\|_{\mathbb{Y}}, \quad \|r\|_{\mathbb{W}} \le C\tau^{1/2}\|F\|_{\mathbb{Y}}.$$

Using $\|D_x^2 r\|_{\mathbb{Y}} \le \|r\|_{\mathbb{W}}$, we obtain

$$\|G_\tau^\pm F\|_{\mathbb{Z}_\tau} \le \|r\|_{\mathbb{Y}} + \tau^{-1}\|\nabla r\|_{\mathbb{Y}} + \tau^{-2}\|r\|_{\mathbb{W}} \le C\Big(\tau^{-1} + \tau^{-3/2} + \tau^{-3/2}\Big)\|F\|_{\mathbb{Y}} \le C\tau^{-1}\|F\|_{\mathbb{Y}}.$$

Thus, we have $\|D_x^2 r\|_{\mathbb{Y}} \le C\|r\|_{\mathbb{W}}$. Combining this with the bound for $\mathcal{M}_\tau^\pm$ yields the desired composition estimate. □

By Lemma 4.2, we can choose τ sufficiently large such that $\|\mathcal{M}_\tau^\pm G_\tau^\pm\|_{\mathbb{Y}\to\mathbb{Y}} \le 1/2 < 1$. Then $I - \mathcal{M}_\tau^\pm G_\tau^\pm$ is an isomorphism on $\mathbb{Y}$. The factorization

$$A_\tau^\pm = B_\tau^\pm - \mathcal{M}_\tau^\pm = (I - \mathcal{M}_\tau^\pm G_\tau^\pm) B_\tau^\pm,$$

implies that $A_\tau^\pm : \mathbb{X}_\pm \to \mathbb{Y}$ are isomorphisms. Their inverses are given by

$$\widetilde{G}_\tau^\pm = G_\tau^\pm \Big(I - \mathcal{M}_\tau^\pm G_\tau^\pm\Big)^{-1},$$

which satisfy the same asymptotic bounds as $G_\tau^\pm$.

Based on the above discussion, we can obtain the following proposition regarding the construction of exponential solutions.

Proposition 4.3. *Fix $x_0 \in \partial\widetilde{\Omega}$, and let ϕ be the associated distance phase satisfying (4.3). Let $\mu_\pm$ be smooth solutions of (4.4) compactly supported in time in $(0,T)$, as constructed above. For all sufficiently large τ, the equations $L_{a,b}^+ v_\tau = 0$ and $L_{a,b}^- p_\tau = 0$ in Q admit solutions of the form*

$$v_\tau = e^{\Theta_\tau}(\mu_+ + R_{+,\tau}), \quad p_\tau = e^{-\Theta_\tau}(\mu_- + R_{-,\tau}),$$

with $v_\tau(0,\cdot) = 0$ and $p_\tau(T,\cdot) = 0$. Here $\Theta_\tau = \tau^2 t + \tau\phi(t,x)$, and the remainders $R_{\pm,\tau} \in \mathbb{X}_\pm$ satisfy (4.5) and the estimate

$$\tau\|R_{\pm,\tau}\|_{\mathbb{Y}} + \tau^{1/2}\|\nabla R_{\pm,\tau}\|_{\mathbb{Y}} + \tau^{-1/2}\Big(\|R_{\pm,\tau}\|_{\mathbb{W}} + \|\partial_\nu R_{\pm,\tau}\|_{\mathbb{N}}\Big) \le C,$$

where $C > 0$ is independent of τ.

Proof. Set $R_{\pm,\tau} = -\tilde{G}_\tau^\pm F_{\pm,\tau}$. Then $R_{\pm,\tau} \in \mathbb{X}_\pm$ and (4.5) holds. The amplitudes $\mu_\pm$ are smooth and independent of τ, so

$$\|P_b^\pm \mu_\pm\|_{\mathbb{Y}} + \|\mu_\pm\|_{\mathbb{Z}_\tau} \le C, \quad \tau \ge 1.$$

Lemma 4.2 therefore gives $\|F_{\pm,\tau}\|_{\mathbb{Y}} \le C$ for all sufficiently large τ. The bounds on $\tilde{G}_\tau^\pm$ yield the stated remainder estimate. The amplitudes vanish near $t = 0$ and $t = T$, and $R_{+,\tau}(0,\cdot) = R_{-,\tau}(T,\cdot) = 0$ by the definition of $\mathbb{X}_\pm$. Hence $v_\tau(0,\cdot) = 0$ and $p_\tau(T,\cdot) = 0$. □

Since $R_{\pm,\tau}|_\Sigma = 0$, the normal traces on Σ reduce to

$$e^{-\Theta_\tau}\partial_\nu v_\tau = \tau(\partial_\nu\phi)\mu_+ + \partial_\nu\mu_+ + \partial_\nu R_{+,\tau},$$
$$e^{\Theta_\tau}\partial_\nu p_\tau = -\tau(\partial_\nu\phi)\mu_- + \partial_\nu\mu_- + \partial_\nu R_{-,\tau}.$$

By Proposition 4.3, the remainder traces satisfy $\|\partial_\nu R_{\pm,\tau}\|_{\mathbb{N}} = O(\tau^{1/2})$.

5. Proof of the main theorem

This section is devoted to proving the main theorem of this paper.

5.1. **Determination of the viscosity.** Our first objective is to recover the metric $g_b(t)$ without assuming prior knowledge of the elastic coefficient in the interior. We begin by establishing an estimate for the Volterra memory terms.

Lemma 5.1. *Let $\phi_j \in C^1([0,T]\times\partial\Omega)$ be real-valued and set $\Theta_j = \tau^2 t + \tau\phi_j$ for $j=1,2$. Let $L = \max_j \|\partial_t\phi_j\|_{L^\infty(\Sigma)}$, $\kappa_\tau = \tau^2 - L\tau$. Suppose that $c \geq 0$ and $E \subset \Sigma$ is a measurable set such that $\phi_2 - \phi_1 \geq c$ on E. Then for all $\tau \geq 2L+1$ and $q \in L^2(\Sigma)$,*

$$\|\mathbf{1}_E e^{-\Theta_2} K_+(e^{\Theta_1} q)\|_{L^2(\Sigma)} \leq e^{-c\tau}\kappa_\tau^{-1}\|q\|_{L^2(\Sigma)}, \tag{5.1}$$

$$\|\mathbf{1}_E e^{\Theta_1} K_-(e^{-\Theta_2} q)\|_{L^2(\Sigma)} \leq e^{-c\tau}\kappa_\tau^{-1}\|q\|_{L^2(\Sigma)}. \tag{5.2}$$

Proof. For $(t,x)\in E$ and $s \leq t$, it follows from the definitions of L and E that

$$\begin{aligned}-\Theta_2(t,x) + \Theta_1(s,x) &= \tau^2(s-t) + \tau\Big(\phi_1(s,x) - \phi_1(t,x)\Big) + \tau\Big(\phi_1(t,x) - \phi_2(t,x)\Big)\\ &\leq -\tau^2(t-s) + \tau L(t-s) - \tau c = -c\tau - \kappa_\tau(t-s).\end{aligned}$$

By Young's convolution inequality, we obtain

$$\begin{aligned}\|\mathbf{1}_E e^{-\Theta_2} K_+(e^{\Theta_1} q)\|_{L^2(\Sigma)} &\leq e^{-c\tau}\left\|\int_0^t e^{-\kappa_\tau(t-s)}|q(s,\cdot)|ds\right\|_{L^2(\Sigma)}\\ &\leq e^{-c\tau}\|e^{-\kappa_\tau(\cdot)}\|_{L^1(0,\infty)}\|q\|_{L^2(\Sigma)}\\ &= e^{-c\tau}\kappa_\tau^{-1}\|q\|_{L^2(\Sigma)}.\end{aligned}$$

Similarly, (5.2) follows by integrating backwards from t to T. □

We now show that the map $\Lambda_{a,b}^T$ uniquely determines b.

Proof of $b_1 = b_2$ in Theorem 1.1. We divide the proof into three steps.

Step 1. Fix $t_0 \in (0,T)$ and define

$$\ell = \max_{x\in\partial\widetilde{\Omega},\, y\in\partial\Omega} |d_2^e(t_0;x,y) - d_1^e(t_0;x,y)|. \tag{5.3}$$

We argue by contradiction, assuming that $\ell > 0$. Interchanging the two coefficient pairs if necessary, we can choose (x_0,y_0) such that $d_2^e(t_0;x_0,y_0) - d_1^e(t_0;x_0,y_0) = \ell$, and define

$$\phi_j(t,x) := d_j^e(t;x_0,x), \quad \Theta_j(t,x) = \tau^2 t + \tau\phi_j(t,x), \quad j=1,2.$$

The metrics $\widetilde{g}_1(t)$ and $\widetilde{g}_2(t)$ agree on $\overline{\widetilde{\Omega}}\setminus\Omega$, and both are simple on $\overline{\widetilde{\Omega}}$. By the connected-intersection property of simple manifolds, every geodesic crossing Ω enters and exits it exactly once, both transversely.

For $j=1,2$, let γ_j be the minimizing geodesic from x_0 to y_0 in the metric $\widetilde{g}_j(t_0)$. Parametrize γ_j by $\widetilde{g}_j(t_0)$-arc length, and let $\xi_j = \widetilde{g}_j(t_0)(\dot\gamma_j(0),\cdot)$ be its initial covector. The first variation formula and the maximality in (5.3) give

$$0 = \frac{\mathrm{d}}{\mathrm{d}s}\Big|_{s=0}\Big(d_2^e(t_0;x(s),y_0) - d_1^e(t_0;x(s),y_0)\Big) = (\xi_1-\xi_2)(w)$$

for any smooth curve $x(s)$ in $\partial\widetilde{\Omega}$ with $x(0)=x_0$ and $\dot x(0)=w$. Thus ξ_1 and ξ_2 have the same restriction to $T_{x_0}\partial\widetilde{\Omega}$. With respect to the same metric at x_0, both covectors have unit norm and positive inward normal components. Their tangential components therefore determine their normal components, so $\xi_1=\xi_2$ and $\dot\gamma_1(0)=\dot\gamma_2(0)$. Since the metrics agree on $\widetilde{\Omega}\setminus\Omega$, uniqueness for the geodesic initial value problem implies that γ_1 and γ_2 coincide up to their first intersection with $\partial\Omega$. Denote this common intersection point by p.

If $p = y_0$, the two geodesics coincide, contradicting $\ell > 0$. Since $p \neq y_0$, uniqueness in the extended simple manifold identifies each segment from p to y_0 with the corresponding

intrinsic geodesic in $\overline{\Omega}$. By strict convexity, these segments lie in Ω between their endpoints and meet Γ transversely at p and y_0.

By transversality at p and smooth dependence on t and x, the geodesics from x_0 to $x \in \Gamma$ remain outside Ω until they reach x for (t, x) near (t_0, p). The metrics agree along these geodesics, so their lengths and terminal covectors agree. The first variation formula then gives $\partial_\nu \phi_1 = \partial_\nu \phi_2$, with a negative common value at (t_0, p). By continuity and the equality $\phi_2(t_0, y_0) - \phi_1(t_0, y_0) = \ell$, we can choose an interval $J \Subset (0, T)$ containing t_0, disjoint open neighborhoods $U_0, W_0 \subset \Gamma$ of p, y_0, respectively, and $c_0 > 0$ such that, for $U = J \times U_0$ and $W = J \times W_0$,

$$\phi_1 = \phi_2 =: \phi, \quad \partial_\nu \phi_1 = \partial_\nu \phi_2 = \partial_\nu \phi < -c_0 \quad \text{on } U, \tag{5.4}$$

$$\phi_2 - \phi_1 \geq \ell/2 \quad \text{on } W. \tag{5.5}$$

Step 2. Let ω_j be the initial Euclidean unit direction of γ_j. Choose smooth, nonnegative transport data for μ_j that are positive at (t_0, ω_j) and compactly supported in a small neighborhood of (t_0, ω_j). By transversality and smooth dependence of the intersections with Γ, we have $\operatorname{supp}(\mu_j|_\Sigma) \subset U \cup W$, $j = 1, 2$. Both amplitudes are positive at (t_0, p), so $\mu_1 \mu_2 > 0$ on a nonempty open subset of U.

Applying Proposition 4.3 to the forward equation for (a_1, b_1) and the backward equation for (a_2, b_2), we can construct the following solutions

$$v_1 = e^{\Theta_1}(\mu_1 + R_1), \quad p_2 = e^{-\Theta_2}(\mu_2 + R_2),$$

with $R_1|_\Sigma = R_2|_\Sigma = 0$. Define q_1 and q_2 by

$$q_1 = \tau \partial_\nu \phi_1 \mu_1 + \partial_\nu \mu_1 + \partial_\nu R_1, \quad q_2 = -\tau \partial_\nu \phi_2 \mu_2 + \partial_\nu \mu_2 + \partial_\nu R_2.$$

We have $\partial_\nu v_1 = e^{\Theta_1} q_1$ and $\partial_\nu p_2 = e^{-\Theta_2} q_2$, and $\|q_1\|_{L^2(\Sigma)} + \|q_2\|_{L^2(\Sigma)} \leq C\tau$. It follows from Proposition 3.1 that $\beta = b_1|_\Sigma = b_2|_\Sigma$ and $\alpha = a_1|_\Gamma = a_2|_\Gamma$. Substituting v_1, p_2 into the boundary identity (2.14), we obtain

$$0 = \mathcal{B}_\tau := \mathcal{B}(v_1, p_2) = \mathcal{L}_\tau(\Sigma) + \mathcal{V}_{+,\tau}(\Sigma) - \mathcal{V}_{-,\tau}(\Sigma) \tag{5.6}$$

where

$$\begin{aligned}
\mathcal{L}_\tau(E) &= \int_E e^{\Theta_1 - \Theta_2} \beta(\mu_2 q_1 - \mu_1 q_2)\, \mathrm{d}S\, \mathrm{d}t, \\
\mathcal{V}_{+,\tau}(E) &= \int_E \alpha \mu_2 e^{-\Theta_2} K_+(e^{\Theta_1} q_1)\, \mathrm{d}S\, \mathrm{d}t, \\
\mathcal{V}_{-,\tau}(E) &= \int_E \alpha \mu_1 e^{\Theta_1} K_-(e^{-\Theta_2} q_2)\, \mathrm{d}S\, \mathrm{d}t.
\end{aligned}$$

Noting that the integrands vanish outside $U \cup W$, we can split the identity into $\mathcal{B}_\tau = \mathcal{B}_\tau(U) + \mathcal{B}_\tau(W)$, where $\mathcal{B}_\tau(E) := \mathcal{L}_\tau(E) + \mathcal{V}_{+,\tau}(E) - \mathcal{V}_{-,\tau}(E)$.

On U, we have $\Theta_1 = \Theta_2$ by (5.4). Since $\partial_\nu \mu_j$ are bounded and $\|\partial_\nu R_j\|_{L^2(\Sigma)} = O(\tau^{1/2})$, we obtain

$$\mathcal{L}_\tau(U) = C_0 \tau + O(\tau^{1/2}), \tag{5.7}$$

where $C_0 = 2 \int_U \beta \partial_\nu \phi\, \mu_1 \mu_2\, dS dt < 0$. Since $\phi_1 = \phi_2$ on U, Lemma 5.1 with $E = U$ and $c = 0$ gives

$$|\mathcal{V}_{+,\tau}(U)| + |\mathcal{V}_{-,\tau}(U)| \leq C\tau^{-1}.$$

Analogously on W, (5.5) and Lemma 5.1 yield

$$|\mathcal{B}_\tau(W)| \leq C\tau e^{-\ell\tau/2}.$$

Combining these estimates, we arrive at the identity

$$0 = C_0 \tau + O(\tau^{1/2}) + O(\tau e^{-\ell\tau/2}). \tag{5.8}$$

For sufficiently large τ, the right-hand side of (5.8) is strictly negative, a contradiction. Hence $\ell = 0$, and

$$d_1^e(t_0; x, y) = d_2^e(t_0; x, y) \quad \text{for all } (x, y) \in \partial\tilde{\Omega} \times \partial\Omega. \tag{5.9}$$

Step 3. Let $y, z \in \partial\Omega$ with $y \neq z$, and extend the intrinsic $g_{b_1}(t_0)$-geodesic from y to z backwards until it meets $\partial\tilde{\Omega}$ at a point x. By the simplicity of $(\widetilde{\Omega}, \widetilde{g}_1(t_0))$, the extended geodesic from x to z and its subsegments are minimizing. The portion from y to z has length $d_1^\Omega(t_0; y, z)$ by construction, so

$$d_1^e(t_0; x, z) = d_1^e(t_0; x, y) + d_1^\Omega(t_0; y, z).$$

On the other hand, the triangle inequality for $\widetilde{g}_2(t_0)$ gives

$$d_2^e(t_0; x, z) \leq d_2^e(t_0; x, y) + d_2^\Omega(t_0; y, z).$$

By (5.9), we obtain $d_1^\Omega(t_0; y, z) \leq d_2^\Omega(t_0; y, z)$. Interchanging the roles of $g_{b_1}(t_0)$ and $g_{b_2}(t_0)$ gives the reverse inequality $d_2^\Omega(t_0; y, z) \leq d_1^\Omega(t_0; y, z)$. Thus the intrinsic boundary distances agree:

$$d_1^\Omega(t_0; y, z) = d_2^\Omega(t_0; y, z) \quad \text{for all } (y, z) \in \partial\Omega \times \partial\Omega.$$

By the boundary rigidity theorem of simple conformally Euclidean manifolds [PSU23, Theorem 11.4.1], we obtain

$$b_1(t_0, x) = b_2(t_0, x) \quad \forall x \in \bar{\Omega}.$$

Since $t_0 \in (0, T)$ is arbitrary and b_j are continuous, it follows that $b_1 = b_2$ on $[0, T] \times \bar{\Omega}$. □

5.2. **Determination of the elastic coefficient.** To complete the proof of Theorem 1.1, it remains to extract the elastic coefficient from the memory term in the mixed identity (2.15). We will assume throughout this section that the conditions of Theorem 1.1 are satisfied and note in view of the last section that it holds:

$$b(t, x) := b_1(t, x) = b_2(t, x), \quad \text{in } Q.$$

We use the integral identity (2.15) to show that, for each $t \in (0, T)$, the scalar geodesic ray transform of $(a_1 - a_2)/b(t, \cdot)$ with respect to $g_b(t)$ vanishes.

To control the τ-dependent remainders, we will show that the rescaled Volterra operators are uniformly bounded on $L^2(Q)$ for large τ and converge strongly to the identity as $\tau \to \infty$.

Lemma 5.2. *Let $\phi \in C^1([0, T] \times \overline{\Omega})$ be real-valued and set $L = \|\phi_t\|_{L^\infty(Q)}$. For $\tau > L$, define*

$$H_\tau F(t, x) = \tau^2 \int_0^t e^{-\tau^2(t-s)+\tau(\phi(s,x)-\phi(t,x))} F(s, x)\, \mathrm{d}s. \tag{5.10}$$

Then

$$\|H_\tau\|_{L^2(Q)\to L^2(Q)} \leq \frac{\tau}{\tau - L} \tag{5.11}$$

and $H_\tau F \to F$ in $L^2(Q)$ for every $F \in L^2(Q)$. The same assertions hold componentwise for vector-valued functions. In particular, if $F_\tau \to F$ in $L^2(Q)$, then $H_\tau F_\tau \to F$ in $L^2(Q)$.

Proof. For $s \leq t$, we have $\phi(s, x) - \phi(t, x) \leq L(t - s)$. Hence, for almost every fixed $x \in \Omega$, Young's convolution inequality in the time variable yields

$$\|H_\tau F(\cdot, x)\|_{L^2(0,T)} \leq \int_0^\infty \tau^2 e^{-(\tau^2-\tau L)r}\, \mathrm{d}r \|F(\cdot, x)\|_{L^2(0,T)} = \frac{\tau}{\tau - L} \|F(\cdot, x)\|_{L^2(0,T)}.$$

Integrating the square of both sides over $x \in \Omega$ and applying Fubini's theorem gives the global bound(5.11). Next, we define the integral operator

$$J_\tau F(t,x) = \tau^2 \int_0^t e^{-\tau^2(t-s)} F(s,x)\,\mathrm{d}s.$$

By extending F by zero outside $(0,T)$, J_τ is identified as a classical one-sided approximate identity. Thus, we have the strong convergence $J_\tau F \to F$ in $L^2(Q)$. A similar convolution argument yields

$$\|H_\tau - J_\tau\|_{L^2\to L^2} \le \int_0^\infty \tau^2 e^{-\tau^2 r}(e^{\tau L r} - 1)\,\mathrm{d}r = \frac{L}{\tau - L}. \tag{5.12}$$

Applying the triangle inequality, we obtain

$$\|H_\tau F - F\|_{L^2} \le \|H_\tau - J_\tau\|_{L^2\to L^2}\|F\|_{L^2} + \|J_\tau F - F\|_{L^2} \to 0.$$

Finally, by the uniform bound (5.11), this strong convergence extends to parameter-dependent inputs $F_\tau \to F$. □

At the end of this section, we are now in a position to complete the proof of Theorem 1.1.

Proof of $a_1 = a_2$ in Theorem 1.1. Write $\delta a = a_1 - a_2$. Fix an arbitrary $t_0 \in (0,T)$ and a nontrivial maximal $g_b(t_0)$-geodesic γ in $\overline{\Omega}$. Since the extensions of b_1 and b_2 also agree outside Ω, write $\widetilde{b}$ for their common value and $\widetilde{g}(t) = \widetilde{b}^{-1}(t,\cdot)e$ for the extended metric. Extend γ backward in the common extended metric $\widetilde{g}(t_0)$ to a point $x_0 \in \partial\widetilde{\Omega}$.

By Proposition 4.3, for this exterior pole x_0 and the associated distance phase function $\phi(t,x)$ satisfying $b|\nabla\phi|^2 = 1$, we can construct exponential solutions v_1 and p_2 solving $L^+_{a_1,b}v_1 = 0$ and $L^-_{a_2,b}p_2 = 0$ respectively, which take the form

$$v_1 = e^{\Theta_\tau}(\mu_+ + R_1), \quad p_2 = e^{-\Theta_\tau}(\mu_- + R_2), \quad \Theta_\tau = \tau^2 t + \tau\phi. \tag{5.13}$$

Substituting these solutions into the mixed volume identity (2.15) in Lemma 2.3 gives

$$\int_Q \delta a\, \nabla(K_+ v_1)\cdot\nabla p_2\,\mathrm{d}x\,\mathrm{d}t = 0. \tag{5.14}$$

Direct computation yields

$$\tau e^{-\Theta_\tau}\nabla(K_+v_1) = H_\tau\left(\nabla\phi(\mu_+ + R_1) + \tau^{-1}\nabla(\mu_+ + R_1)\right),$$
$$\tau^{-1}e^{\Theta_\tau}\nabla p_2 = -\nabla\phi(\mu_- + R_2) + \tau^{-1}\nabla(\mu_- + R_2).$$

By the remainder estimates in Proposition 4.3 and Lemma 5.2, we have

$$\tau e^{-\Theta_\tau}\nabla(K_+v_1) \to \mu_+\nabla\phi, \quad \tau^{-1}e^{\Theta_\tau}\nabla p_2 \to -\mu_-\nabla\phi$$

strongly in $L^2(Q)$ as $\tau\to\infty$. Taking the limit as $\tau\to\infty$ in (5.14) and using the eikonal equation, we obtain

$$\int_Q \frac{\delta a}{b}\,\mu_+\mu_-\,\mathrm{d}x\,\mathrm{d}t = 0. \tag{5.15}$$

It remains to remove the transport weight. Let (r,ω) be the geodesic polar coordinates centered at x_0, with the volume element $dx = J_e(t,r,\omega)\,\mathrm{d}r\,\mathrm{d}\omega$. We note that along the geodesic $\gamma_{t,x_0,\omega}$, the differential operators can be expressed as

$$\widetilde{b}\nabla\phi\cdot\nabla = \partial_r, \quad \mathrm{div}_e(\widetilde{b}\nabla\phi) = \partial_r \log J_e.$$

Then the transport equations (4.4) can be rewritten as

$$2\partial_r\mu_\pm + \left(\partial_r \log J_e \mp \phi_t\right)\mu_\pm = 0.$$

Multiplying the transport equations for μ_+ and μ_- by μ_- and μ_+, respectively, and adding the resulting equations gives

$$\partial_r(\mu_+\mu_-) + (\partial_r \log J_e)\mu_+\mu_- = 0,$$

which implies

$$\partial_r\Big(J_e\mu_+\mu_-\Big) = 0. \tag{5.16}$$

Let $I \Subset (0,T)$ be an interval containing t_0, and let V be a small neighborhood of the initial direction of the extended geodesic at x_0. Choose $r_* > 0$ so that the initial segment $0 < r \le r_*$ lies in $\widetilde{\Omega} \setminus \overline{\Omega}$ for all $(t,\omega) \in I \times V$. After shrinking I and V, let $r_-(t,\omega)$ and $r_+(t,\omega)$ be the entry and exit arc-length parameters of the ray in Ω. The crossings are transverse, so these parameters depend smoothly on (t,ω) by the implicit function theorem. For arbitrary $\chi_\pm \in C_c^\infty(I \times V)$, set

$$\mu_\pm(t, r_*, \omega) = J_e(t, r_*, \omega)^{-1/2}\chi_\pm(t,\omega).$$

Thus $J_e\mu_+\mu_- = \chi_+\chi_-$ along the ray from r_* through Ω. Substituting the product into (5.15) gives

$$\int_{I\times V} \chi_+\chi_-\Big(\int_{r_-(t,\omega)}^{r_+(t,\omega)} \frac{\delta a(\gamma_{t,x_0,\omega}(r))}{b(t,\gamma_{t,x_0,\omega}(r))}dr\Big)\,\mathrm{d}\omega\,\mathrm{d}t = 0. \tag{5.17}$$

For any $\chi_+ \in C_c^\infty(I \times V)$, choose $\chi_- \in C_c^\infty(I \times V)$ equal to one on a neighborhood of $\operatorname{supp}\chi_+$. Equation (5.17) then implies that the inner integral vanishes as a distribution on $I \times V$. By continuity in (t,ω), it vanishes pointwise. In particular, the integral along γ at time t_0 is zero. Since γ was an arbitrary nonconstant maximal $g_b(t_0)$-geodesic, we obtain

$$I_{g_b(t_0)}\left(\frac{\delta a}{b(t_0,\cdot)}\right) = 0.$$

The manifold $(\overline{\Omega}, g_b(t_0))$ is simple and has dimension $n \ge 2$. Injectivity of the scalar geodesic ray transform [PSU23, Theorem 4.7.7] therefore gives $\delta a/b(t_0,\cdot) = 0$. Since $b(t_0,\cdot) > 0$, we conclude that $\delta a = 0$. Therefore the proof of Theorem 1.1 is complete. □

6. Numerical reconstruction

In this section, we present some numerical experiments of reconstructing the elastic coefficient $a(x)$ and the time-dependent viscosity $b(t,x)$ by time-domain full-waveform inversion. For each choice of boundary displacement, we solve the Kelvin–Voigt equation and compute moments of the resulting total traction. This gives a reduced optimization problem in the material coefficients. We refer to [Ple06] for the adjoint-state approach and to [PKN16] for related viscoelastic waveform inversion.

6.1. Traction data and reduced objective.

Let h_ℓ be a boundary velocity, $f_\ell = K_+h_\ell$ the associated displacement, and k_m a boundary test function. The corresponding traction moment is given by

$$F_{\ell m}(a,b) = \int_\Sigma (\Lambda_{a,b}^T f_\ell)k_m\,\mathrm{d}S\,\mathrm{d}t. \tag{6.1}$$

The boundary input determines the traction response, and the test function determines how we weight that response in space and time. For some inputs, we gradually change the boundary displacement and then keep it fixed. We include moments of the traction measured while the displacement is fixed in the fit.

We use the positive parametrization

$$a_\theta = a_{\mathrm{ref}} e^{A_\theta}, \quad b_\theta = b_{\mathrm{ref}} e^{B_\theta}, \quad A_\theta = \sum_{i=1}^{265} \theta_i^a \psi_i, \quad B_\theta = \sum_{j=1}^{48} \theta_j^b \Psi_j. \tag{6.2}$$

Here $a_{\mathrm{ref}} = 1$ and $b_{\mathrm{ref}} = 0.4$. The functions ψ_i are local tensor-product cubic B-splines restricted to the disk. The viscosity basis consists of the six spatial polynomials $1, x, y, x^2 - y^2, 2xy, x^2 + y^2 - \frac{1}{2}$ multiplied by the first eight Legendre polynomials in $2t/T - 1$. This extends the temporal expansion from degree four to degree seven.

Let F_{i} and F_{d} be the moment maps on the inversion and data grids. For the full spline fit, we initialize at the coarse-grid estimate θ_{p} and evaluate the grid correction there. We use a regularization penalty $\mathcal{R}(\theta)$ on the coefficient parameters, specified in the next subsection, with weight $\lambda > 0$. Given the observations Y^δ, the residual and the reduced objective are

$$\begin{aligned} c &= F_{\mathrm{d}}(\theta_{\mathrm{p}}) - F_{\mathrm{i}}(\theta_{\mathrm{p}}), \quad r(\theta) = W \operatorname{vec}\Big(F_{\mathrm{i}}(\theta) + c - Y^\delta\Big), \\ \mathcal{J}_\lambda(\theta) &= \tfrac{1}{2}\|r(\theta)\|_2^2 + \lambda \mathcal{R}(\theta). \end{aligned} \tag{6.3}$$

We keep the correction c fixed throughout each continuation run. For a group of L_g inputs, we multiply row ℓ of the residual by $\sqrt{1+\delta^2}/(\sqrt{2L_g}\,\|Y^\delta_{\ell,:}\|_2)$. These weights account for the size of each observed row and give equal weight in the fit to the two groups: observations over the full time window and observations taken while the boundary displacement is fixed. Let $v_\ell = \partial_t u_\ell$ and let p_m solve the terminal adjoint equation $L^-_{a,b} p_m = 0$ with boundary value k_m and zero terminal data. Applying (2.13) to the linearized state gives

$$DF_{\ell m}(a,b)[\delta a, \delta b] = \int_Q (\delta a \nabla u_\ell + \delta b \nabla v_\ell) \cdot \nabla p_m \, \mathrm{d}x \, \mathrm{d}t. \tag{6.4}$$

The explicit variation of the traction cancels the boundary term from the linearized divergence source. For parameter variations we substitute $\delta a = a_\theta \delta A_\theta$ and $\delta b = b_\theta \delta B_\theta$ in (6.4).

6.2. **Joint updates and choice of regularization parameter.** In the full spline fit, $\mathcal{R}$ includes a quadratic amplitude term and a pseudo-Huber gradient penalty on A_θ, a squared gradient penalty on a_θ, and a weighted squared Hessian penalty on A_θ. The Hessian weight is $(1 + |\nabla A_{\theta_{\mathrm{p}}}|^2/3^2)^{-1}$, which allows steeper transitions near the contours of the initial fit. The viscosity penalty is quadratic in B_θ, its spatial gradient, and its time derivative. Restricted elastic updates use directions derived from smoothed coarse-grid estimates. These directions describe contour shifts, contrast changes, and low-order background variations. Each restricted fit retains all 48 viscosity parameters and uses a quadratic penalty in the normalized reduced coordinates.

At each iterate θ_n, we use the discrete adjoint to compute the gradient g_n of the reduced objective. The damped direction solves a quadratic subproblem with objective

$$g_n^T d + \tfrac{1}{2} d^T \Big(H_n + \lambda R_n + \mu_n D_n\Big) d. \tag{6.5}$$

Here H_n is the Gauss–Newton data curvature or its secant update, R_n is a positive approximation of the penalty curvature, and D_n is a positive diagonal scaling. We retain the cross block between the elastic and viscosity parameters and solve for both increments at the same step. The full fit constrains $\theta_n + d$ to $[-1.2, 1.2]^{313}$. The restricted fit uses directions in the reduced space and accepts trial coefficients in $[-1.6, 1.6]^{313}$. We adjust the damping parameter according to the ratio of the actual decrease in the objective to the decrease predicted by the local model. We stop when the norm of the gradient restricted to the admissible directions meets the prescribed tolerance.

We solve the inverse problem for selected values of $\lambda_j = 1.5 \times 10^{-4} 3^{-j}$ in decreasing order, using each fit to initialize the next. The discrepancy test accounts for noise and a grid allowance estimated from c in (6.3). We compare stationary fits satisfying the discrepancy test on reserved waveforms from both groups. For noiseless data, a weighted difference between predictions on two grids sets the comparison threshold. For noisy data, we favor stronger regularization when score differences are within the estimated noise variation.

6.3. **Discretization and experimental setting.** We take Ω to be the unit disk and $T = 3$. The spatial meshes triangulate the same inscribed regular 64-gon and use continuous piecewise quadratic finite elements. The data grid has 3329 nodes and 384 time steps. Five selected fits use the full spline space on a separate grid with 1777 nodes and 288 time steps. The restricted fit for the annulus at 2% noise uses 39 elastic update directions, 865 nodes, and 224 time steps.

We apply the implicit midpoint rule to the displacement and velocity. Let M, A, and $\mathsf{B}_{n-1/2}$ be the mass and stiffness matrices for unit density, a_θ, and $b_\theta(t_{n-1/2}, \cdot)$. The displacement increment is $\Delta t\, \mathbf{v}^{n-1/2}$, and the nodal reaction is

$$\mathbf{T}^{n-1/2} = \frac{\mathsf{M}(\mathbf{v}^n - \mathbf{v}^{n-1})}{\Delta t} + \mathsf{B}_{n-1/2}\mathbf{v}^{n-1/2} + \mathsf{A}\mathbf{u}^{n-1/2}. \tag{6.6}$$

The superscript $n-1/2$ denotes the average of the two endpoint values for the states. The reaction vanishes at the interior nodes. We approximate (6.1) by pairing the boundary residual with the nodal test values and summing in time. Linearizing the time-stepping and traction equations gives the discrete tangent, whose transpose is the discrete adjoint.

The full-window group consists of 55 boundary inputs with 336 moments per input. The tests pair 21 real Fourier modes of orders 0 through 10 with 16 smooth time windows centered at equally spaced times from 0.075 to 2.925. The additional group consists of 20 displacement ramps. For each ramp input, we retain 126 traction moments from the last six time windows, all within the interval of fixed boundary displacement. For validation, we reserve seven inputs with observations over the full time window and sixteen with observations taken while the displacement is fixed. For each observed row of clean data $Y_{\ell,:}$, we draw an independent standard Gaussian vector ξ_ℓ of the corresponding length and set

$$Y^\delta_{\ell,:} = Y_{\ell,:} + \delta \|Y_{\ell,:}\|_2 \frac{\xi_\ell}{\|\xi_\ell\|_2}, \quad \delta \in \{0, 0.02\}. \tag{6.7}$$

We use one noise realization per case at 2%.

6.4. **Reconstruction results.** The three elastic profiles contain two elliptic inclusions, a rotated kite, and a five-lobed annulus, respectively. The corresponding viscosity fields vary in amplitude, orientation, and location, respectively. All target fields and all six final reconstructions are strictly positive on their respective domains. The spline partition of unity bounds A_θ between the smallest and largest elastic coordinates. The spatial factors and the Legendre polynomials are uniformly bounded on their domains, which bounds B_θ. These bounds and (6.2) give positive lower bounds for the two reconstructed fields.

Figures 1–3 show the truth and the reconstructions from noiseless and single-acquisition 2% data. The two inclusions remain separated in Case 1. The reconstructions in Case 2 recover the overall kite outline, with local amplitude variations near its tips. In Case 3, the outer lobes and central low-value region are visible in both reconstructions. The viscosity slices show the change in amplitude in Case 1, rotation in Case 2, and motion of the high-value region in Case 3.

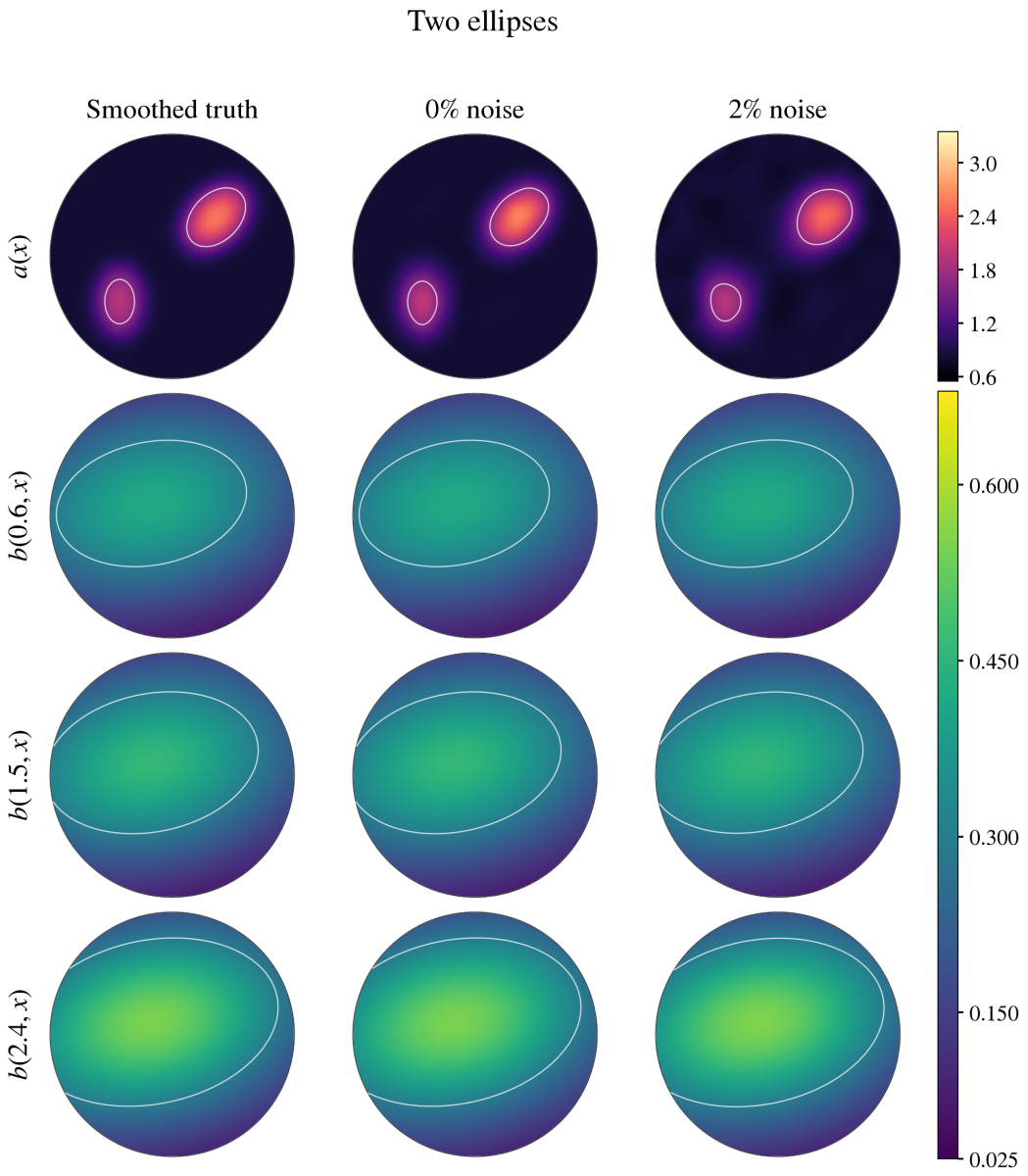


FIGURE 1. Case 1. The elastic field has two elliptic inclusions. The columns show the truth and the reconstructions from noiseless and 2% noisy data. The upper row shows $a(x)$ and the remaining rows show $b(t, x)$ at $t = 0.6, 1.5, 2.4$. Each field uses a common color scale. The contours mark $a = 1.65$ and $b = 0.30$.

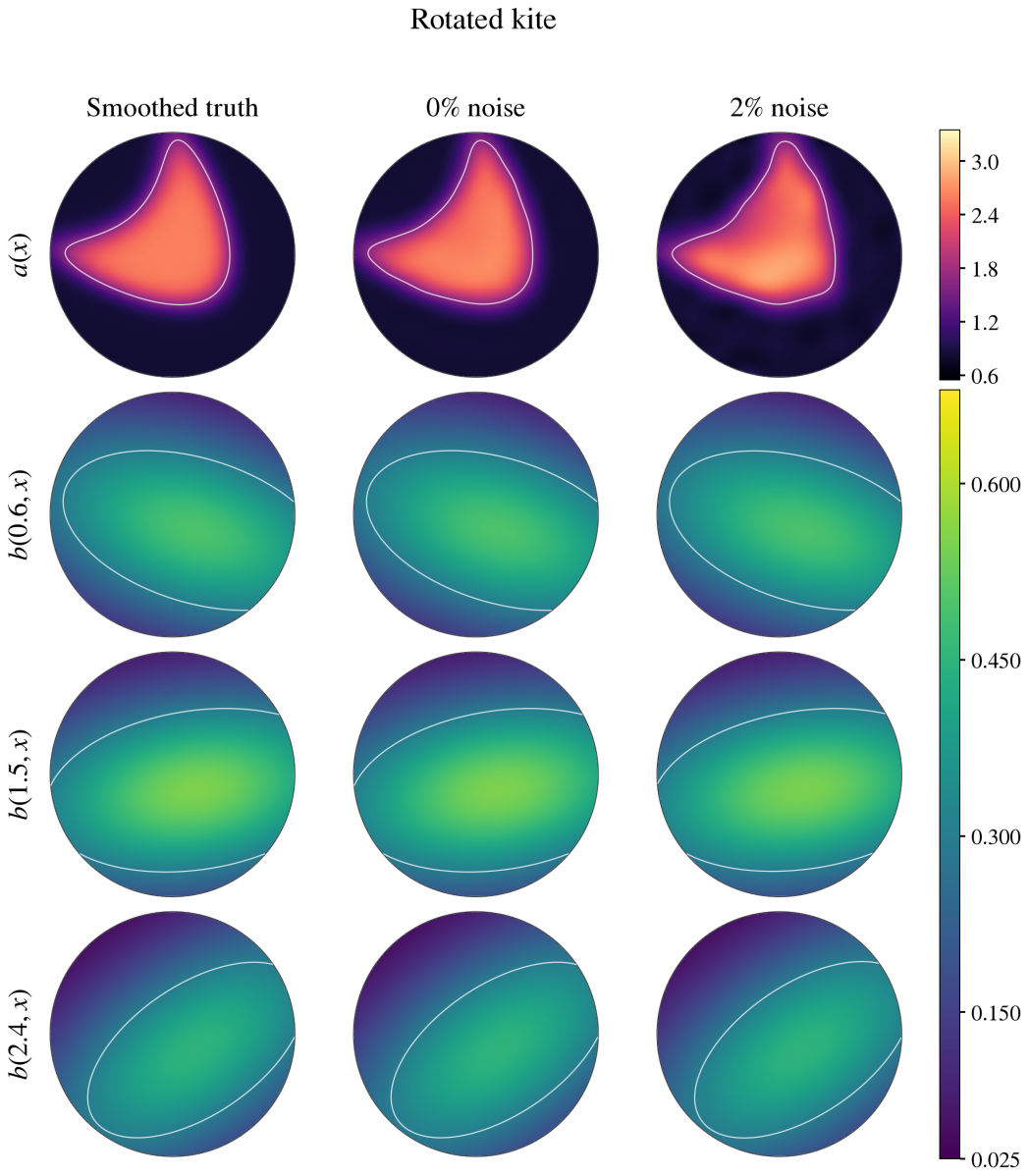


FIGURE 2. Case 2. Reconstruction of the rotated kite and the viscosity with a rotating principal axis. The columns, time slices, and color scales follow Figure 1.

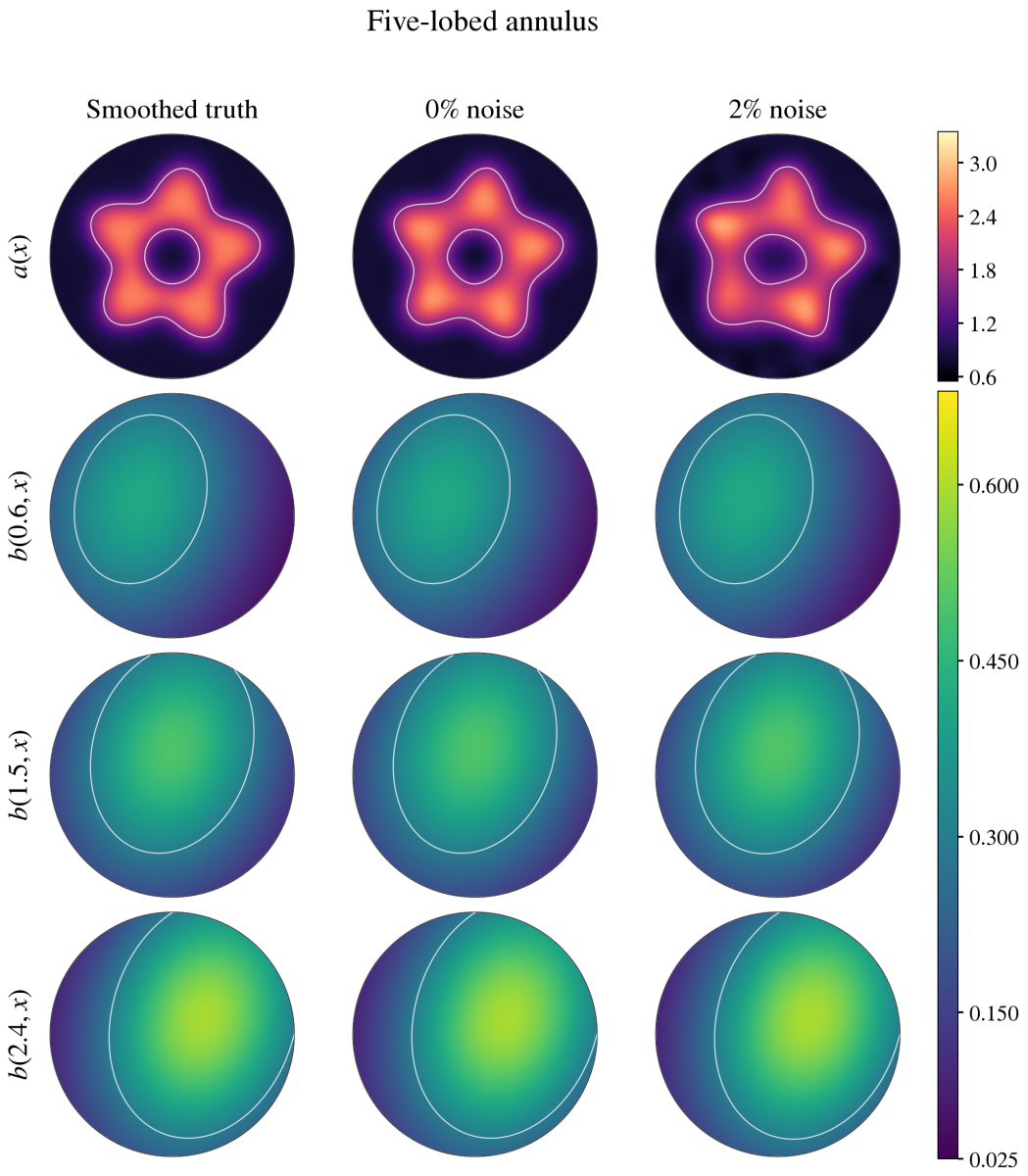


FIGURE 3. Case 3. Reconstruction of the five-lobed annulus and the viscosity with a moving center. The columns, time slices, and color scales follow Figure 1.

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Data availability statement. The numerical experiments are based on synthetic data generated using the models and parameters described in this article. The data supporting the numerical results are available from the corresponding author upon reasonable request.

Conflict of Interests. Hereby we declare there are no conflicts of interest.

References

[AP26] S. Acosta and B. Palacios. Simultaneous determination of wave speed, diffusivity, and nonlinearity in the Westervelt equation using complex time-periodic solutions. *SIAM J. Appl. Math.*, 86(1):1–20, 2026.

[BDI97] A. L. Bukhgeim, G. V. Dyatlov, and V. M. Isakov. Stability of memory reconstruction from the Dirichlet-to-Neumann operator. *Siberian Math. J.*, 38(4):636–646, 1997.

[BDU02] A. L. Bukhgeim, G. V. Dyatlov, and G. Uhlmann. Reconstruction of the memory from partial boundary measurements. In *Mathematical Results in Quantum Mechanics*, volume 307 of *Contemporary Mathematics*, pages 39–46. American Mathematical Society, 2002.

[BSB08] N. Baranibalan, K. Sakthivel, and K. Balachandran. Uniqueness and stability in inverse parabolic equations with memory. *Nonlinear Anal. Hybrid Syst.*, 2(4):1077–1088, 2008.

[CdHS26] G. Covi, M. de Hoop, and M. Salo. Propagation of singularities and inverse problems for the viscoacoustic wave equation, 2026. arXiv:2603.24497 [math.AP].

[CG09] F. Colombo and D. Guidetti. Identification of the memory kernel in the strongly damped wave equation by a flux condition. *Commun. Pure Appl. Anal.*, 8(2):601–620, 2009.

[Chr82] R. M. Christensen. *Theory of Viscoelasticity: An Introduction*. Academic Press, New York, second edition, 1982.

[CK01] B. Canuto and O. Kavian. Determining coefficients in a class of heat equations via boundary measurements. *SIAM J. Math. Anal.*, 32(5):963–986, 2001.

[Col07] F. Colombo. An inverse problem for the strongly damped wave equation with memory. *Nonlinearity*, 20(3):659–683, 2007.

[DHP07] R. Denk, M. Hieber, and J. Prüss. Optimal L^p-L^q-estimates for parabolic boundary value problems with inhomogeneous data. *Math. Z.*, 257(1):193–224, 2007.

[EKAG23] H. Ebrahimi, M. Krsmanovic, H. Ali, and R. Ghosh. Material-geometry interplay in damping of biomimetic scale beams. *Appl. Phys. Lett.*, 123(8):081701, 2023.

[Fei24] A. Feizmohammadi. An inverse boundary value problem for isotropic nonautonomous heat flows. *Math. Ann.*, 388(2):1569–1607, 2024.

[GS26] H. Grebnev and P. Stefanov. High frequency wave propagation for the viscoelastic wave equation with singular memory, 2026. arXiv:2608.30138 [math.AP].

[HKXG21] J.-T. Hang, Y. Kang, G.-K. Xu, and H. Gao. A hierarchical cellular structural model to unravel the universal power-law rheological behavior of living cells. *Nat. Commun.*, 12(1):6067, 2021.

[IY19] O. Yu. Imanuvilov and M. Yamamoto. Carleman estimate and an inverse source problem for the Kelvin–Voigt model for viscoelasticity. *Inverse Problems*, 35(12):125001, 2019.

[KRU26] Y. Kian, Z. Rezig, and G. Uhlmann. Inverse coefficient problem for the wave equation with nonlocal attenuation, 2026. arXiv:2609.19854 [math.AP].

[LZ24a] L. Li and Y. Zhang. Inverse problems for a quasilinear strongly damped wave equation arising in nonlinear acoustics. *J. Differential Equations*, 410:786–831, 2024.

[LZ24b] L. Li and Y. Zhang. On inverse problems for a strongly damped wave equation on compact manifolds. *J. Geom. Anal.*, 34(4):118, 2024.

[PKN16] A. Pakravan, J. W. Kang, and C. M. Newtson. A Gauss–Newton full-waveform inversion for material profile reconstruction in viscoelastic semi-infinite solid media. *Inverse Probl. Sci. Eng.*, 24(3):393–421, 2016.

[Ple06] R.-E. Plessix. A review of the adjoint-state method for computing the gradient of a functional with geophysical applications. *Geophys. J. Int.*, 167(2):495–503, 2006.

[PSU23] G. P. Paternain, M. Salo, and G. Uhlmann. *Geometric Inverse Problems: With Emphasis on Two Dimensions*, volume 204 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2023.

[SO20] G. Seriani and S. P. Oliveira. Numerical modeling of mechanical wave propagation. *Riv. Nuovo Cim.*, 43:459–514, 2020.

[Wu12] B. Wu. Carleman estimate for a strongly damped wave equation and applications to an inverse problem. *Math. Methods Appl. Sci.*, 35(4):427–437, 2012.

[YZZ26] M. Yamamoto, W. Zhang, and Z. Zhao. Stability of coefficients inverse problem of 3D viscoelastic system. *Inverse Probl. Imaging*, 24:53–80, 2026.

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